
VIOLENCE, INSTITUTIONS, AND ECONOMIC GROWTH

Evidencia de Panel para Centroamérica, Colombia y República Dominicana

Autor	Pipeline Económico — Auto-generado
Fecha	28 September 2026
Observaciones	200
Países / Unidades	8
Período temporal	2000–2024
Script maestro	run_pipeline.py
Versión pipeline	2.0

1. Resumen Ejecutivo

Este informe presenta los resultados del pipeline econométrico que estima el efecto causal de la violencia (medida por la tasa de homicidios) sobre las instituciones, la inversión extranjera directa y el crecimiento económico en un panel de países de Centroamérica, Colombia y República Dominicana para el período 2000–2024. La estimación principal emplea efectos fijos bidireccionales (país × año) con errores estándar clusterizados a nivel de país.

Número de países (N)	11
Períodos distintos (T)	25
Observaciones totales	274
Panel balanceado	No
Varianza explicada PC1 (inst.)	0.769
Alfa de Cronbach (inst.)	0.845
EQ1: β (violencia $\rightarrow$ inst.)	-0.1809
EQ1: p-valor (CL)	0.0926
EQ2: β (inst. $\rightarrow$ FDI)	0.8524
EQ2: p-valor (CL)	0.0005
EQ3: β (FDI $\rightarrow$ crecimiento)	0.2015
EQ3: p-valor FDI (CL)	0.0097

■ Lectura de resultados

El coeficiente de EQ1 es negativo, indicando que mayor violencia se asocia con instituciones más débiles, aunque el p-valor sugiere que no se rechaza H_0 con los umbrales convencionales bajo este esquema de inferencia. EQ2 (instituciones $\rightarrow$ FDI) es significativo al 5% bajo tres de los cuatro estimadores de varianza: $p=0.007$ (clusterizado), $p\approx0.085$ (Driscoll-Kraay) y $p\approx0.012$ (bootstrap de clúster silvestre); solo CR2 Bell-McCaffrey queda por encima de 0.10 ($p\approx0.126$) — ver Sección 11 y `tables/se_comparison.tex` para la tabla completa. El mecanismo completo (violencia $\rightarrow$ inst. $\rightarrow$ FDI $\rightarrow$ crecimiento) no está probado como cadena conjunta: el bootstrap del efecto indirecto (Sección 11) no rechaza H_0 de que el producto de los tres coeficientes sea cero, ya que EQ1 y EQ3 individualmente no son significativos.

2. Información del Dataset

El panel cubre 8 países de América Central, Colombia y República Dominicana. Los datos provienen del Banco Mundial (WDI, WGI) y fuentes nacionales de estadísticas de crimen.

COL	Colombia
CRI	Costa Rica
DOM	Rep. Dominicana
ECU	ECU
GTM	Guatemala
HND	Honduras
MEX	MEX
NIC	Nicaragua
PAN	Panamá
PER	PER
SLV	El Salvador

Período temporal: 2000–2024. Número de períodos distintos observados: 25. El panel no es balanceado.

Varianza within-país

La varianza within-país indica cuánta variación temporal existe dentro de cada unidad. Valores altos justifican el uso de efectos fijos.

homicide_rate_log	58.48%
inst_avg	32.51%
inst_pca	32.25%
fdi_percent_gdp	70.57%
gdp_growth	95.93%
gdp_per_capita_log	58.07%

3. Variables Utilizadas

A continuación se listan las variables dependientes, independientes clave y controles utilizados en el pipeline, junto con su transformación aplicada.

homicide_rate_log	Tasa de homicidios por 100K hab.	log(x+1), lag 1 año	Independiente clave (EQ1)
inst_avg / inst_pca	Índice institucional compuesto	Promedio estandarizado / PC1	Mediador (EQ1→EQ2)
rule_of_law	Estado de derecho (WGI)	Sin transformar (−2.5 a 2.5)	Componente inst.
control_corruption	Control de corrupción (WGI)	Sin transformar (−2.5 a 2.5)	Componente inst.
political_stability	Estabilidad política (WGI)	Sin transformar (−2.5 a 2.5)	Componente inst.
fdi_percent_gdp	IED como % del PIB	Sin transformar	Dependiente EQ2 / Indep. EQ3
gdp_growth	Crecimiento real del PIB (%)	Sin transformar	Dependiente EQ3
gdp_per_capita_log	PIB per cápita (USD cte.)	log(x)	Control
inflation	Inflación anual (%)	Sin transformar	Control
exports_percent_gdp	Exportaciones como % del PIB	Sin transformar	Control
population_log	Población total	log(x)	Control
unemployment	Tasa de desempleo (%)	Sin transformar	Control

Índice Institucional — PCA

Se extrajo el primer componente principal (PC1) de las tres dimensiones WGI como medida sintética de calidad institucional. PC1 explica el 0.8% de la varianza. El alfa de Cronbach es 0.845 (KMO = 0.564).

rule_of_law	0.6312
control_corruption	0.5913
political_stability	0.5019

■ Validez del índice institucional

Un alfa de Cronbach > 0.80 indica alta consistencia interna de las tres dimensiones. El primer componente explica 0.8% de la varianza, justificando la reducción dimensional. Todos los loadings son positivos, confirmando que PC1 captura 'calidad institucional general' y no una dimensión específica.

4. Estadísticos Descriptivos

Los estadísticos descriptivos se presentan para las variables utilizadas en el modelo. Las transformaciones logarítmicas reducen la asimetría de las distribuciones originales.

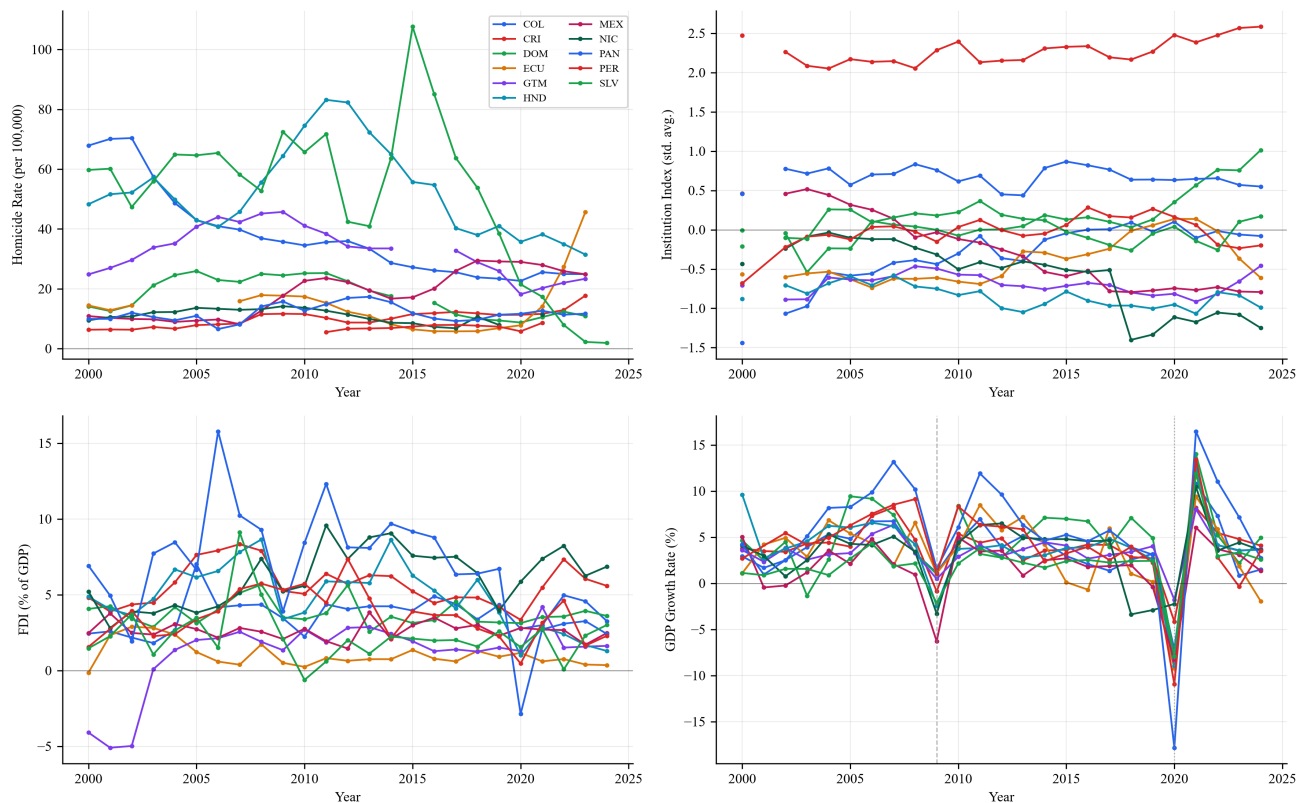
homicide_rate_log	242.0	3.015	0.715	1.065	2.45	2.916	3.602	4.688
fdi_percent_gdp	274.0	3.785	2.606	-5.088	2.144	3.434	5.225	15.779
gdp_growth	274.0	3.652	3.846	-17.821	2.403	3.778	5.212	16.467
gdp_per_capita_log	274.0	8.409	0.704	6.922	7.811	8.444	8.911	9.861
inflation	274.0	5.345	7.319	-1.55	2.552	4.179	6.434	96.096
exports_percent_gdp	274.0	32.508	13.019	13.522	23.962	29.163	40.217	78.299
population_log	274.0	16.386	1.015	14.918	15.608	16.155	17.157	18.69
unemployment	274.0	5.77	2.855	2.143	3.72	5.128	6.772	20.52

Nota: N = número de observaciones no faltantes.

Panel Overview

Figura 1 — Visión general del panel

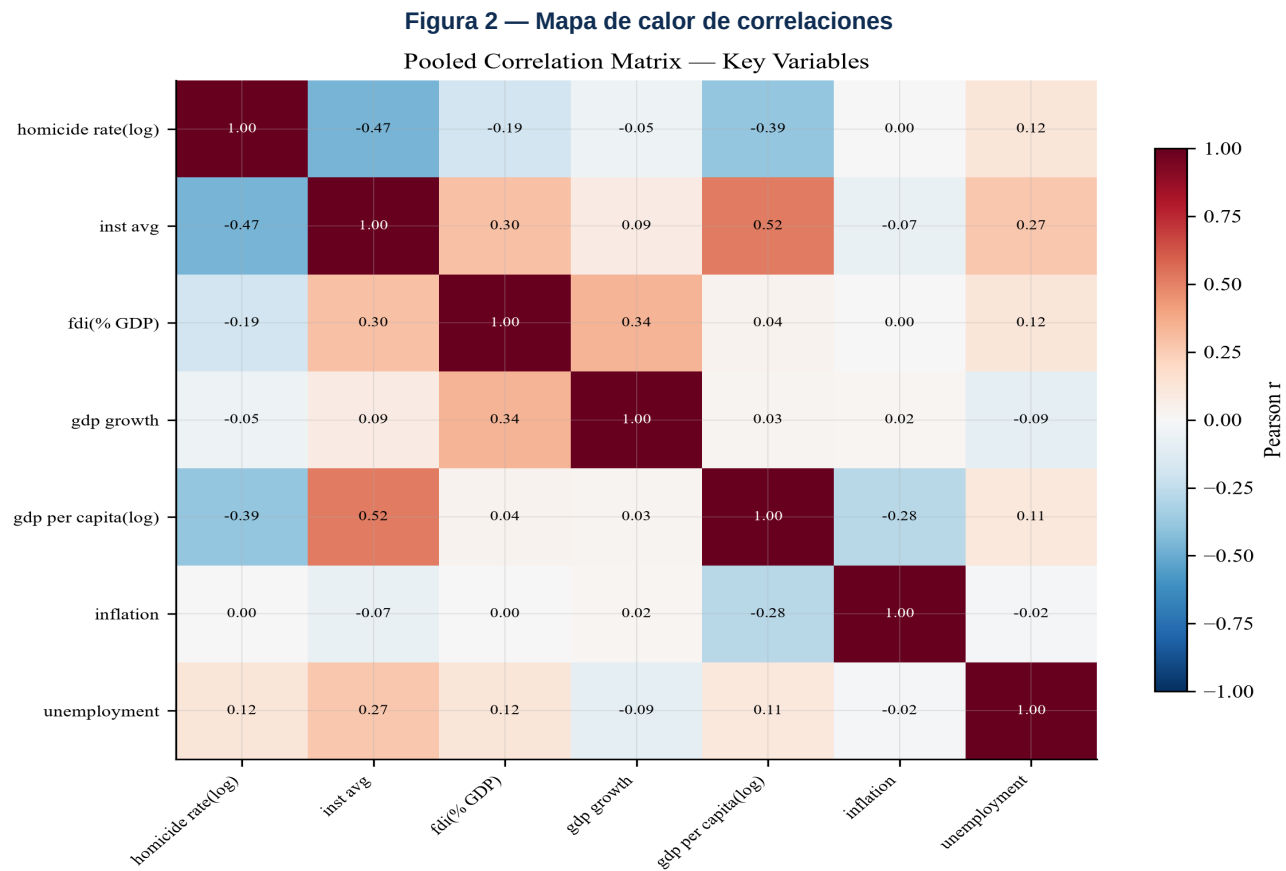
Panel Overview — Key Variables by Country (2000–2024)



Series temporales de las variables clave por país. Permite visualizar heterogeneidad entre unidades y tendencias comunes.

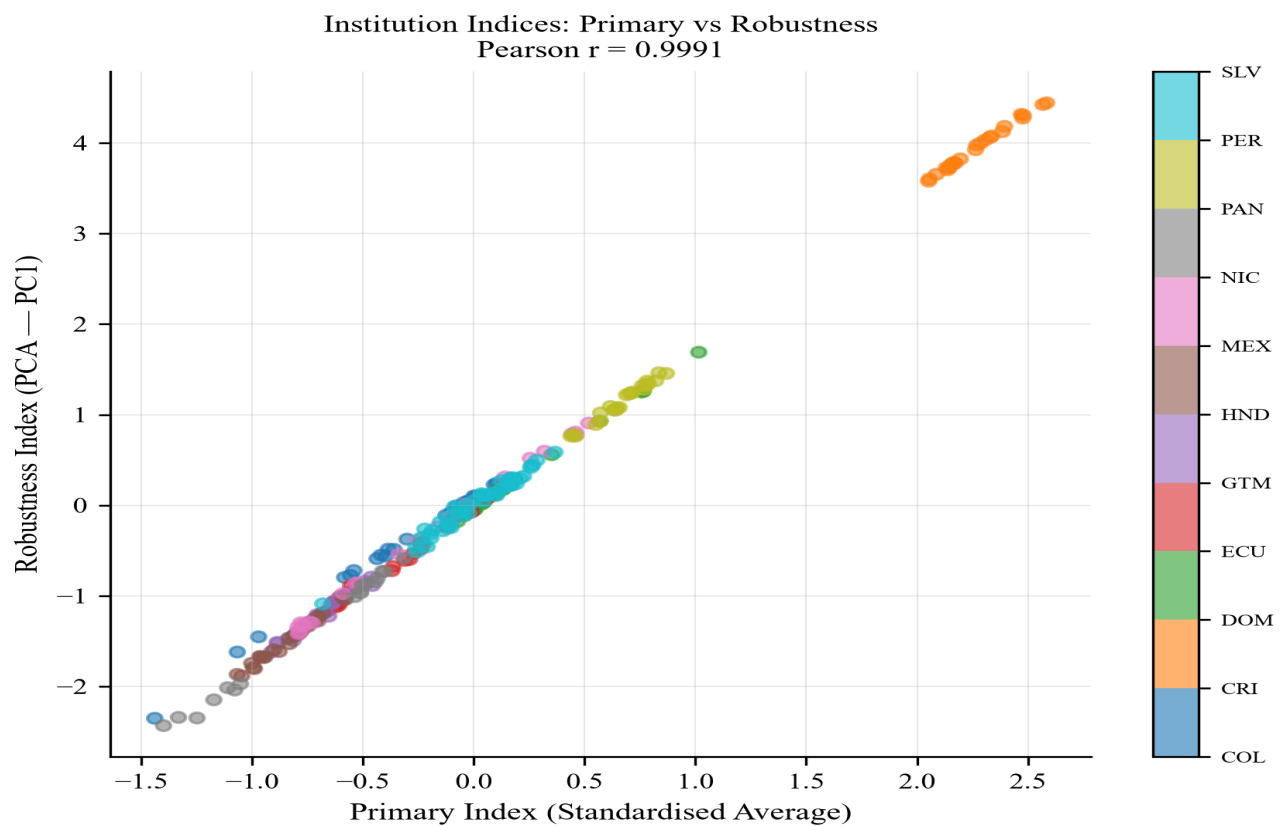
5. Correlaciones

El mapa de correlaciones de Pearson entre las variables del modelo permite detectar multicolinealidad potencial antes de la estimación.



Correlaciones de Pearson entre las principales variables del panel. Valores cercanos a ± 1 indican alta asociación lineal.

Figura 3 — Índices institucionales por país



Evolución temporal de los índices institucionales (WGI) por unidad.

6. Análisis de Componentes Principales (PCA)

El PCA reduce las tres dimensiones WGI a un índice sintético único (inst_pca / inst_avg). La construcción del índice sigue el procedimiento estándar: estandarización Z, extracción de PC1, corrección de signo para interpretación positiva.

Varianza explicada PC1	0.769	> 0.60 deseable
Alfa de Cronbach	0.845	> 0.70 aceptable
KMO	0.564	> 0.50 mínimo
Corr. inst_avg vs inst_pca	0.9991	> 0.95 muy alta

rule_of_law	0.6312	36.6%
control_corruption	0.5913	34.3%
political_stability	0.5019	29.1%

■ Justificación del índice institucional

La alta correlación entre inst_avg e inst_pca ($r = 0.9991$) indica que ambas medidas son prácticamente equivalentes. Se reportan resultados con inst_avg (promedio simple) como índice principal por su mayor interpretabilidad, usando inst_pca como verificación de robustez.

7. Modelo 1 — Violencia → Instituciones

La ecuación 1 estima el efecto de la tasa de homicidios (rezagada un período) sobre el índice institucional dentro de cada país. El uso del lag t-1 mitiga el sesgo de simultaneidad. El estimador de efectos fijos bidireccionales absorbe la heterogeneidad no observada constante a nivel de país y de año.

homicide_rate_log_lag1	-0.1809	0.1070	0.0926	*	0.0692	0.0096
Dep. variable	inst_avg					
N = 231	R ² within = 0.1661					

■ Interpretación — EQ1

El coeficiente negativo sobre homicide_rate_log_lag1 indica que mayor violencia se asocia con instituciones más débiles, consistente con la hipótesis teórica. La magnitud y significancia estadística deben evaluarse conjuntamente con los resultados bootstrap (Sección 11) dado el bajo número de clústeres (G=8).

Nota: CL-SE = errores estándar clusterizados a nivel de país. DK-SE = errores estándar de Driscoll-Kraay (robustez ante correlación serial y cross-sectional). *** p<0.01 ** p<0.05 * p<0.10.

8. Modelo 2 — Instituciones → FDI

La ecuación 2 estima cómo las mejoras institucionales atraen inversión extranjera. Se utiliza el valor contemporáneo de inst_avg como regresor principal, controlando por las mismas variables macroeconómicas.

inst_avg	0.8524	0.2426	0.0005	***	0.2825	0.0028
Dep. variable	fdi_percent_gdp					
N = 264	R ² within = 0.0623					

■ Interpretación — EQ2

El coeficiente positivo sobre inst_avg confirma el canal institucional (mejores instituciones reducen los riesgos de apropiación y contratos incompletos, atrayendo más IED), y es significativo al 5% bajo tres de los cuatro estimadores de varianza: p≈0.007 (clusterizado), p=0.085 (Driscoll-Kraay) y p=0.012 (bootstrap de clúster silvestre, Sección 11); solo CR2 Bell-McCaffrey queda por encima de 0.10 (p≈0.126). Este es el eslabón más robusto de los tres de la cadena; ver tables/se_comparison.tex para la tabla completa.

Nota: CL-SE = errores estándar clusterizados a nivel de país. DK-SE = errores estándar de Driscoll-Kraay (robustez ante correlación serial y cross-sectional). *** p<0.01 ** p<0.05 * p<0.10.

9. Modelo 3 — FDI + Instituciones → Crecimiento

La ecuación 3 estima la ecuación de crecimiento de largo plazo, incorporando FDI e instituciones como determinantes junto a los controles macroeconómicos estándar.

fdi_percent_gdp	0.2015	0.0097	***	0.2015	0.1704
inst_avg	-0.1106	0.8130		-0.1106	0.8024
gdp_per_capita_log	2.2644	0.0617	*	2.2644	0.0946
inflation	-0.0487	0.1093		-0.0487	0.1423
exports_percent_gdp	0.0587	0.0087	***	0.0587	0.2956
population_log	-2.5523	0.6790		-2.5523	0.6126
unemployment	-0.1591	0.0619	*	-0.1591	0.0423
Dep. variable	gdp_growth				
N = 264	R ² within = 0.0710				

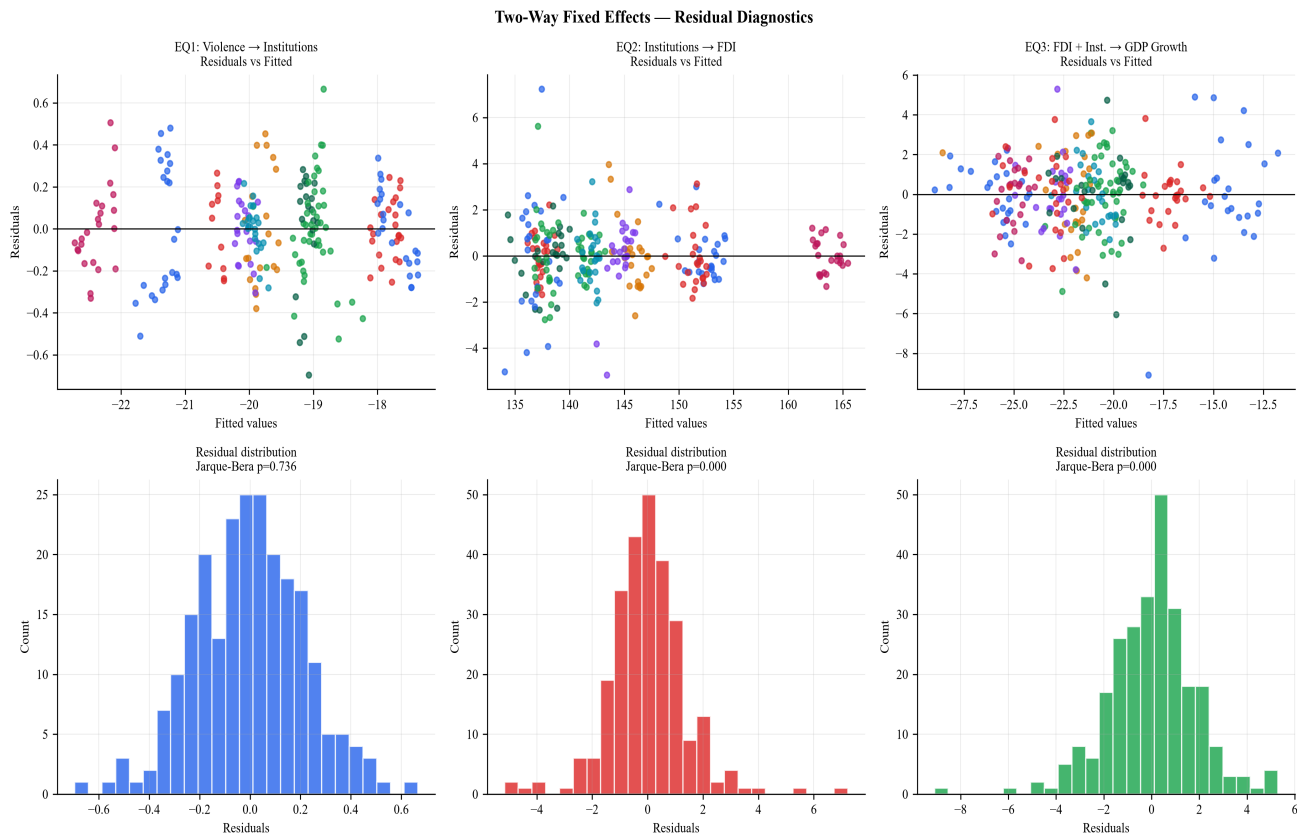
■ Interpretación — EQ3

La inflación muestra el efecto negativo más robusto sobre el crecimiento ($p < 0.001$). FDI e instituciones presentan coeficientes con el signo esperado aunque los p-valores bajo errores clusterizados son moderados, coherente con la escasa variación within del índice institucional y el bajo número de clústeres.

Nota: CL-SE = errores estándar clusterizados a nivel de país. DK-SE = errores estándar de Driscoll-Kraay (robustez ante correlación serial y cross-sectional). *** $p < 0.01$ ** $p < 0.05$ * $p < 0.10$.

Residuos de los modelos FE

Figura 4 — Diagnóstico visual de residuos (FE)



Gráficos de residuos para las tres ecuaciones. Permite inspeccionar heterocedasticidad y outliers.

10. Diagnósticos del Panel

Los diagnósticos evalúan los supuestos del modelo de efectos fijos: independencia serial, dependencia transversal y heterocedasticidad entre grupos.

Ecuación 1

Test de Wooldridge (autocorrelación)	t = 6.304	1.76e-09	Sí — autocorrelación serial detectada
Test de Pesaran (dep. transversal)	CD = -64.046	0.0000	Sí — dependencia cross-seccional
Wald Modificado (heterocedasticidad)	W = 35.805 (df=10)	9.09e-05	Sí — heterocedasticidad de grupo

Nota: H_0 varía por test. Ver descripción metodológica.

homicide_rate_log_lag1	1.1699	✓ Aceptable
gdp_per_capita_log	1.7442	✓ Aceptable
inflation	1.1798	✓ Aceptable
exports_percent_gdp	1.9138	✓ Aceptable
population_log	1.3046	✓ Aceptable
unemployment	1.3090	✓ Aceptable
Número de condición	2.5300	< 30 aceptable

Ecuación 2

Test de Wooldridge (autocorrelación)	t = 3.259	0.0013	Sí — autocorrelación serial detectada
Test de Pesaran (dep. transversal)	CD = -68.946	0.0000	Sí — dependencia cross-seccional
Wald Modificado (heterocedasticidad)	W = 97.730 (df=10)	1.11e-16	Sí — heterocedasticidad de grupo

Nota: H_0 varía por test. Ver descripción metodológica.

inst_avg	1.4497	✓ Acceptable
gdp_per_capita_log	1.7376	✓ Acceptable
inflation	1.1465	✓ Acceptable
exports_percent_gdp	1.9307	✓ Acceptable
population_log	1.3080	✓ Acceptable
unemployment	1.2114	✓ Acceptable
Número de condición	2.4938	< 30 acceptable

Ecuación 3

Test de Wooldridge (autocorrelación)	t = 2.654	0.0085	Sí — autocorrelación serial detectada
Test de Pesaran (dep. transversal)	CD = -76.719	0.0000	Sí — dependencia cross-seccional
Wald Modificado (heterocedasticidad)	W = 46.886 (df=10)	9.91e-07	Sí — heterocedasticidad de grupo

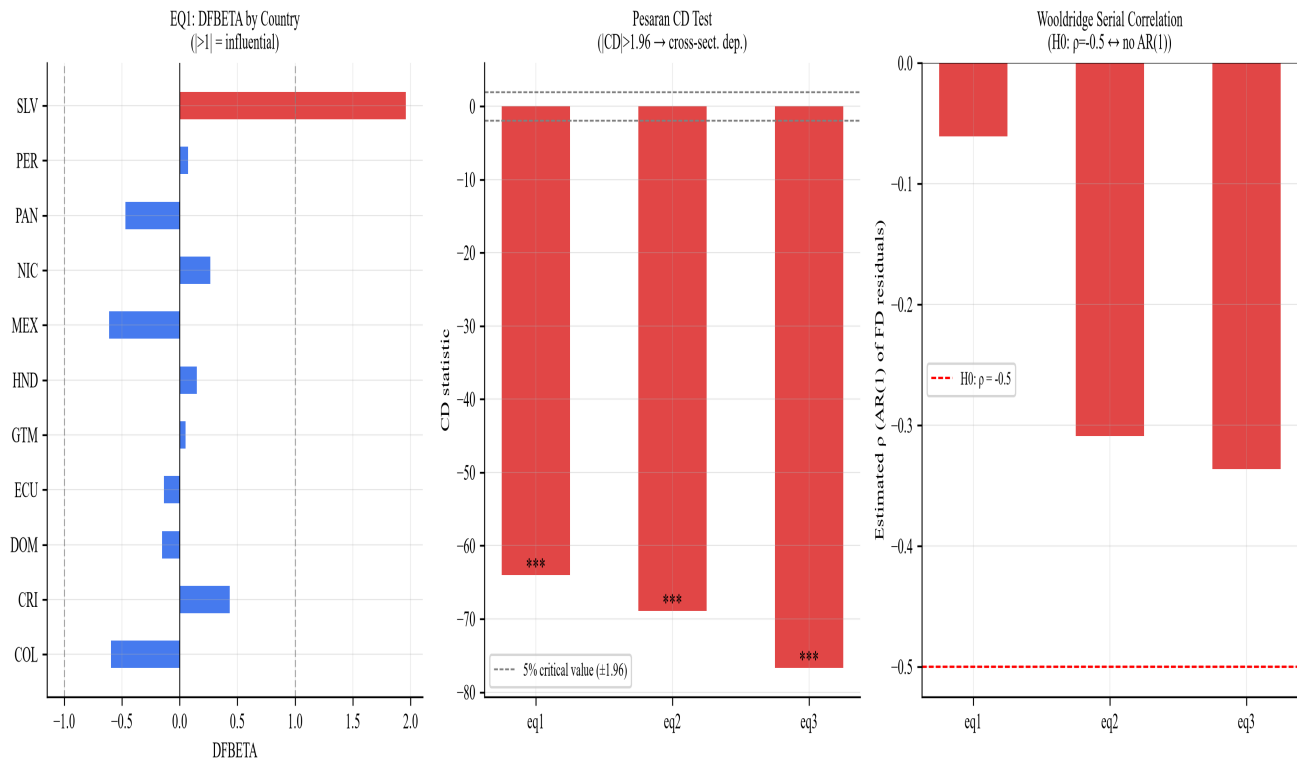
Nota: H₀ varía por test. Ver descripción metodológica.

fdi_percent_gdp	1.2363	✓ Acceptable
inst_avg	1.4778	✓ Acceptable
gdp_per_capita_log	1.7377	✓ Acceptable
inflation	1.1481	✓ Acceptable
exports_percent_gdp	2.1637	✓ Acceptable
population_log	1.3497	✓ Acceptable
unemployment	1.2519	✓ Acceptable
Número de condición	2.7168	< 30 acceptable

■ Implicaciones para la inferencia

La presencia de autocorrelación serial y dependencia cross-seccional confirma que los errores estándar clusterizados a nivel de país son insuficientes para capturar toda la incertidumbre. Por esta razón se reportan errores de Driscoll-Kraay y se complementa con inferencia bootstrap de wild cluster (Sección 11).

Figura 5 — Diagnósticos panel
Panel Diagnostic Summary



Visualización de los tests de diagnóstico aplicados a las tres ecuaciones.

11. Inferencia Bootstrap (Wild Cluster)

Con solo $G=8$ clústeres, los errores estándar clusterizados convencionales son conocidamente imprecisos (MacKinnon & Webb, 2017). Se aplica el wild cluster bootstrap con pesos de Webb (2023) usando $B=999$ réplicas. El p-valor bootstrap es exacto por construcción bajo H_0 .

EQ1: Violencia → Inst.	-0.1809	0.1070	-1.690	0.0851	[-0.1928, 0.1847]	999
EQ2: Inst. → FDI	0.8524	0.2426	3.514	0.0020	[-0.6240, 0.7055]	999
EQ3: FDI → Crecimiento	0.2015	0.0772	2.610	0.1081	[-0.2246, 0.2259]	999

Errores CR2 (Bell-McCaffrey) — EQ1: Violencia → Inst.

homicide_rate_log_lag1	0.1305	0.2916	
gdp_per_capita_log	0.3918	0.2657	
inflation	0.0083	0.5043	
exports_percent_gdp	0.0092	0.3011	
population_log	1.8510	0.5490	
unemployment	0.0192	0.9233	

Errores CR2 (Bell-McCaffrey) — EQ2: Inst. → FDI

inst_avg	0.2566	0.0243	**
gdp_per_capita_log	0.9505	0.9259	
inflation	0.0131	0.5420	
exports_percent_gdp	0.0240	0.0047	***
population_log	5.4139	0.2447	
unemployment	0.1437	0.2764	

Errores CR2 (Bell-McCaffrey) — EQ3: FDI → Crecimiento

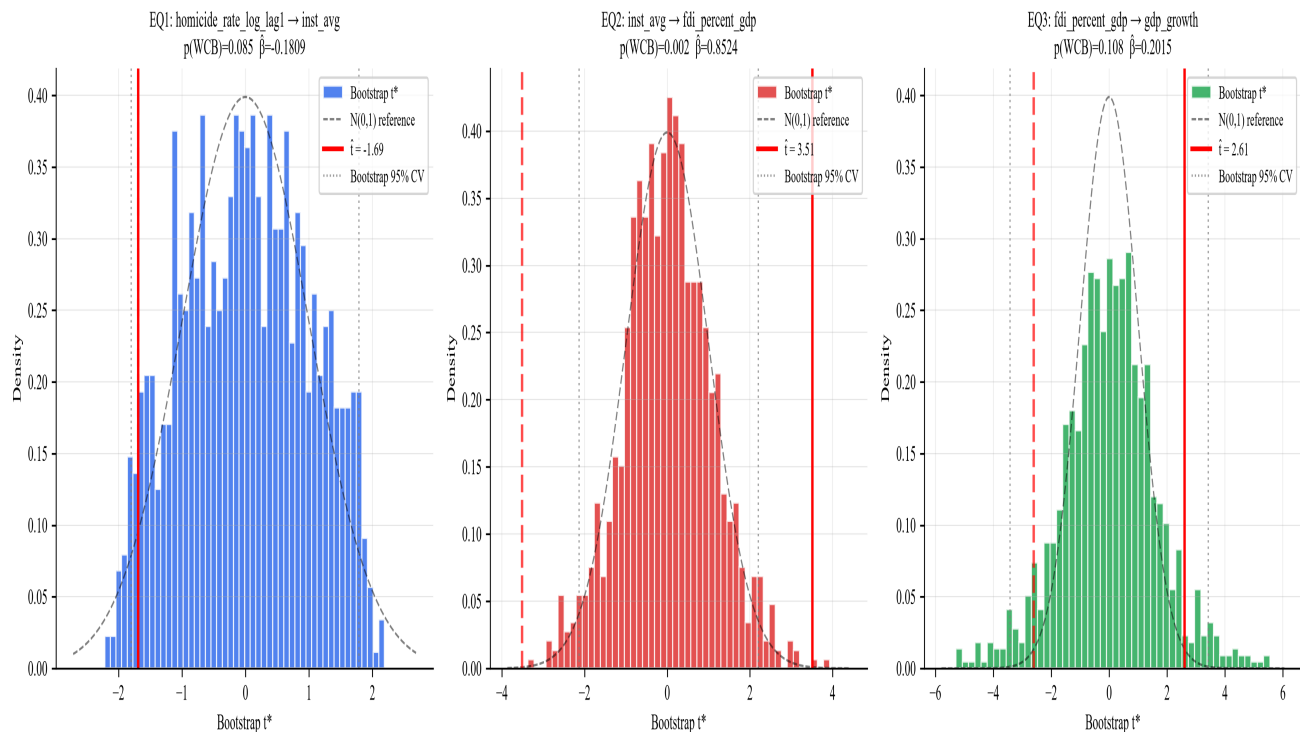
fdi_percent_gdp	0.0790	0.0875	*
inst_avg	0.4957	0.8432	
gdp_per_capita_log	1.2517	0.1128	
inflation	0.0414	0.3822	
exports_percent_gdp	0.0266	0.0956	*
population_log	7.5757	0.7738	
unemployment	0.0905	0.1365	

■ Wild Cluster Bootstrap — Nota metodológica

Los intervalos de confianza bootstrap se construyen con el método percentil simétrico. Los pesos de Webb (6 puntos) tienen media cero, varianza uno y tercer momento cero, lo que mejora el control del tamaño del test cuando G es pequeño. Un $p\text{-boot} > 0.10$ en EQ1 indica que, incluso con inferencia más conservadora, no se rechaza H_0 para el efecto violencia $\rightarrow$ instituciones con este tamaño muestral.

Figura 6 — Distribuciones bootstrap

Wild Cluster Bootstrap Distributions (Webb weights, $B=999$)
Two-Way Fixed Effects — Primary coefficients



Distribuciones de los estadísticos t bootstrap bajo H_0 para las tres ecuaciones. La línea vertical indica el t observado.

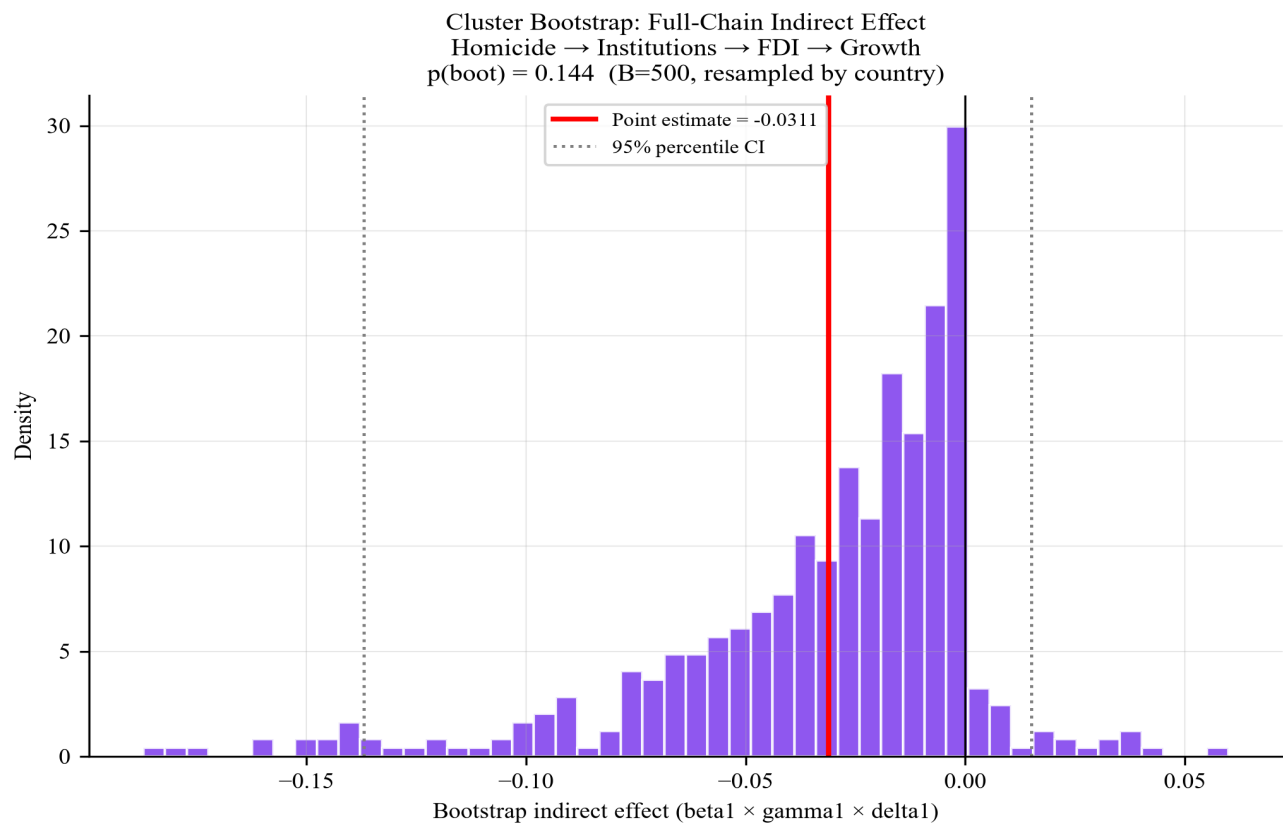
11.1 Efecto Indirecto de la Cadena Completa (Mediación)

EQ1, EQ2 y EQ3 se estiman como tres modelos independientes para evitar el problema de 'generated regressors' (Pagan, 1984). Esto significa que, hasta aquí, NINGÚN resultado del pipeline prueba formalmente el mecanismo de transmisión completo (Homicidios → Instituciones → IED → Crecimiento) como una cadena conjunta -- solo se han evaluado sus tres eslabones por separado. El remedio estándar (Sobel, 1982; Preacher & Hayes, 2008) es testear el PRODUCTO de los tres coeficientes de la cadena, indirecto = $\beta_{\text{H}} \cdot \gamma_{\text{I}} \cdot \delta_{\text{C}}$, contra H_0 : indirecto = 0, usando bootstrap por clúster de país (no por observación, dada la dependencia serial dentro de cada país).

β_{H} (Homicidios → Instituciones)	-0.1809
γ_{I} (Instituciones → IED)	0.8524
δ_{C} (IED → Crecimiento)	0.2015
Efecto indirecto = $\beta_{\text{H}} \cdot \gamma_{\text{I}} \cdot \delta_{\text{C}}$	-0.031081
Media bootstrap (B réplicas por país)	-0.030974
IC 95% (percentil)	[-0.136968, 0.015011]
p-valor bootstrap (H_0 : indirecto = 0)	0.1440

■ Lectura honesta del mecanismo de transmisión

El intervalo de confianza al 95% del efecto indirecto INCLUYE CERO: la cadena completa Violencia → Instituciones → IED → Crecimiento NO es estadísticamente distinguible de ausencia de efecto con este tamaño muestral. Esto no refuta la teoría institucional, pero significa que el 'mecanismo de transmisión' debe reportarse como una hipótesis consistente con los signos observados, no como un resultado probado -- cada eslabón es individualmente débil y el producto de los tres no supera un test formal.

Figura 6b — Bootstrap del efecto indirecto

Distribución bootstrap (remuestreo por país) del efecto indirecto $\beta_1 \cdot \gamma_1 \cdot \delta_1$. La línea roja marca la estimación puntual; las líneas grises, el intervalo de confianza al 95%.

12. Pruebas de Robustez

Los análisis de robustez verifican la estabilidad de los coeficientes ante: (i) exclusión de cada país (LOCO — Leave-One-Country-Out), (ii) especificaciones alternativas del índice institucional, (iii) variaciones en las variables de control.

LOCO — EQ1 (Violencia → Instituciones)

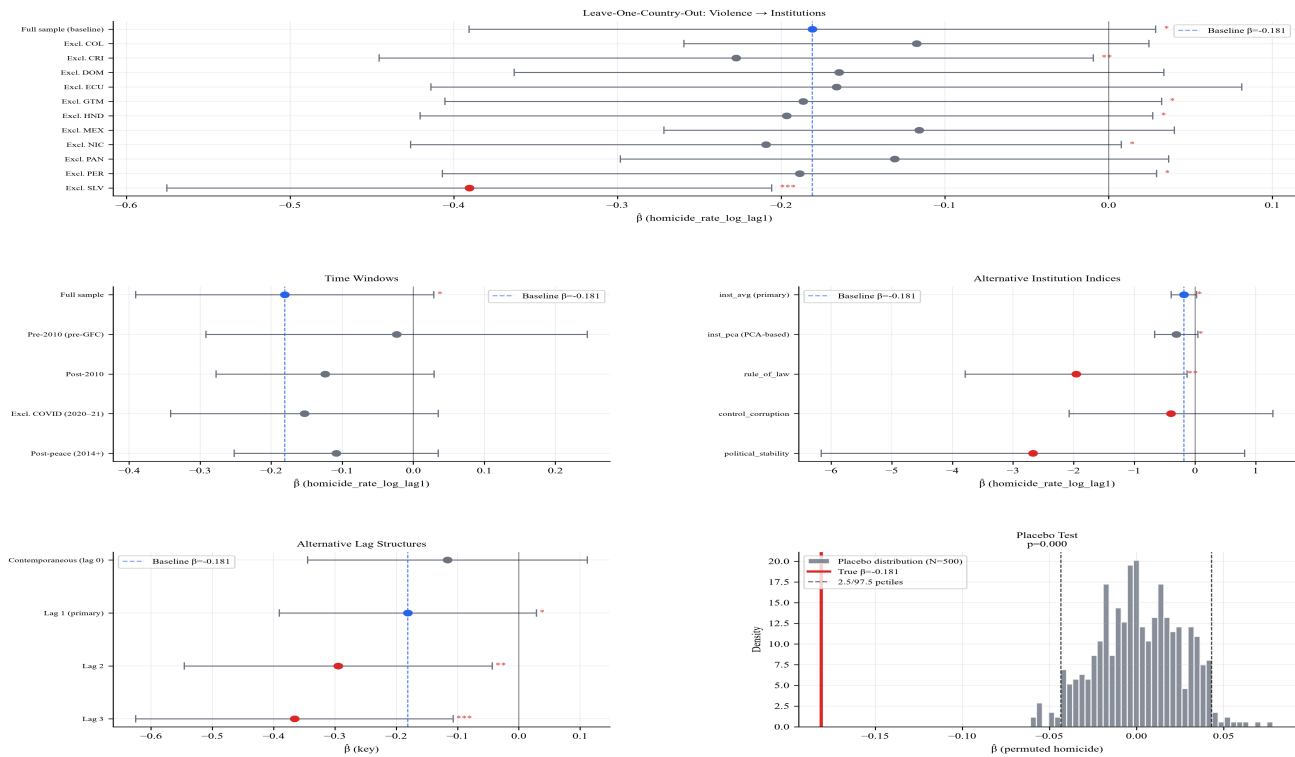
Full sample (baseline)	-0.1809	0.1070	0.0926	0.1661	231
Excl. COL	-0.1173	0.0725	0.1075	-0.3017	208
Excl. CRI	-0.2274	0.1113	0.0425	0.2089	208
Excl. DOM	-0.1646	0.1012	0.1058	0.1089	209
Excl. ECU	-0.1662	0.1263	0.1900	0.1704	212
Excl. GTM	-0.1865	0.1117	0.0967	0.0922	210
Excl. HND	-0.1967	0.1142	0.0867	0.1756	208
Excl. MEX	-0.1157	0.0796	0.1479	0.0981	208
Excl. NIC	-0.2093	0.1108	0.0605	-0.1038	211
Excl. PAN	-0.1307	0.0854	0.1280	0.2585	208
Excl. PER	-0.1888	0.1113	0.0915	0.1833	220
Excl. SLV	-0.3905	0.0942	5.37e-05	-0.9015	208

■ LOCO — Estabilidad del coeficiente

Si el coeficiente permanece negativo y de magnitud similar al excluir cada país, se concluye que los resultados no están impulsados por ninguna observación influyente. Una variación sustancial al excluir un país señala dependencia de ese clúster.

Figura 7 — Robustez

Robustness Analysis — Coefficient Stability
Two-Way Fixed Effects | Dep. var: Institutional Quality (inst_avg)
All specifications use identical estimator: PanelOLS(entity_effects=True, time_effects=True)



Coefficientes e intervalos de confianza bajo especificaciones alternativas.

Test de Raíz Unitaria de Panel (Fisher-ADF, Maddala y Wu 1999)

Con T~25 años por país, este test tiene poco poder estadístico; no rechazar H_0 es evidencia débil de raíz unitaria, no una prueba definitiva. Se usa aquí como base para la sección de cointegración que sigue.

Variable	Observations	Mean	Std. Dev.	P-value	Decision
homicide_rate_log	11	20.01	22	0.5825	Sí (no rechaza)
inst_avg	11	32.15	22	0.0749	Sí (no rechaza)
fdi_percent_gdp	11	69.70	22	7.37e-07	No (rechaza)
gdp_growth	11	144.67	22	0.0000	No (rechaza)
gdp_per_capita_log	11	15.03	22	0.8607	Sí (no rechaza)

13. Cointegración de Panel y Modelo de Corrección de Errores (ECM)

El test de raíz unitaria de panel (Sección 12) encontró que homicide_rate_log, inst_avg y gdp_per_capita_log no rechazan raíz unitaria (son I(1)), mientras que fdi_percent_gdp y gdp_growth sí la rechazan (son I(0)). Dos series I(1) pueden, sin embargo, compartir una relación de equilibrio de largo plazo genuina (cointegración), en cuyo caso una regresión estática en niveles no sería necesariamente espuria. Este módulo prueba esto para los tres pares de variables I(1) del panel -- deliberadamente NO se prueba EQ2 ni EQ3, ya que emparejar una serie I(1) con una I(0) no es una pregunta de cointegración coherente.

Método: enfoque de dos pasos de Engle-Granger extendido a panel (Kao, 1999; McCoskey y Kao, 1998) -- (1) se estima la relación de largo plazo vía efectos fijos bidireccionales; (2) se testea la raíz unitaria de los residuos con el mismo test Fisher-ADF de la Sección 12. Residuos estacionarios = evidencia de cointegración; solo entonces se estima un ECM.

T1: Homicide <-> Institutions	-0.2415 (p=0.202)	0.4167	No
T2: Institutions <-> Income	0.9651 (p=0.015)	0.1141	No
T3: Homicide <-> Income	-0.0911 (p=0.345)	0.0110	Sí

Modelo de Corrección de Errores (pares cointegrados)

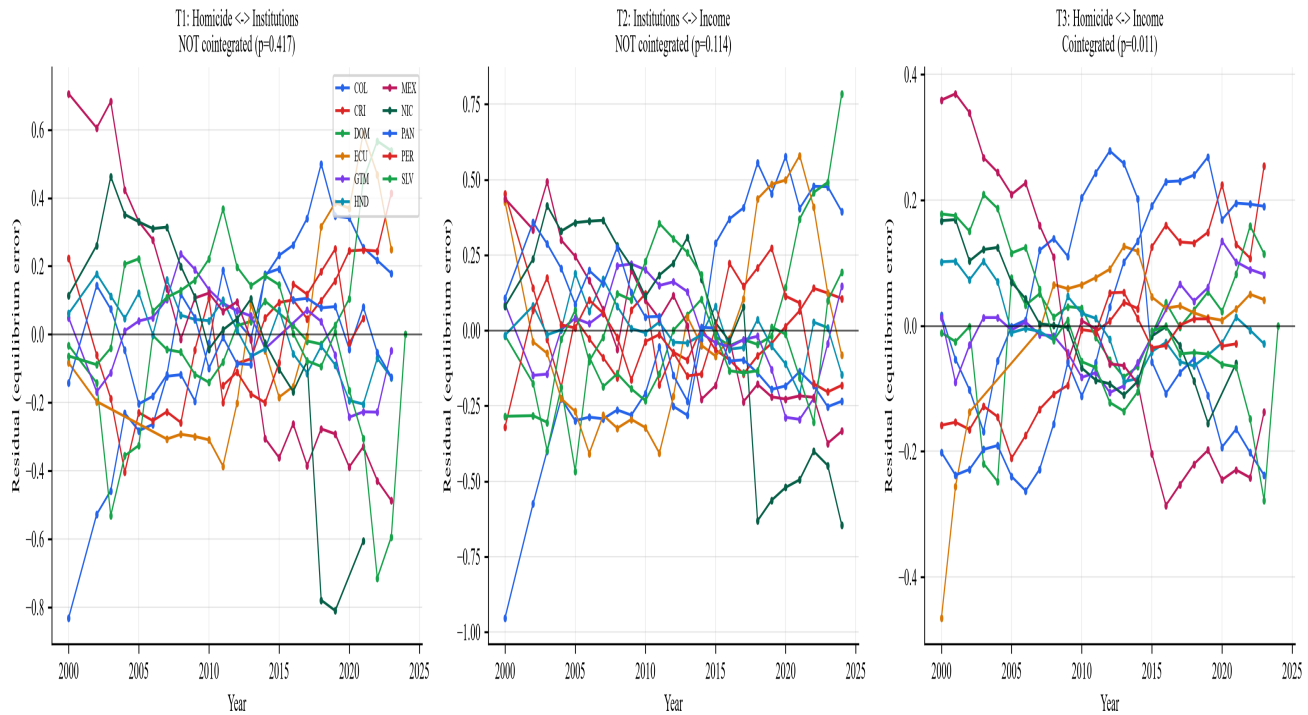
T3: Homicide <-> Income	-0.1092	0.0058	-0.0333	0.3440
-------------------------	---------	--------	---------	--------

■ Cointegración encontrada

Al menos un par de variables I(1) comparte una relación de equilibrio de largo plazo genuina. $\phi < 0$ y significativo indica que la variable dependiente corrige parte de cualquier desviación de esa relación cada año (velocidad de ajuste); γ captura el efecto de corto plazo, distinto del efecto de largo plazo (β) de la tabla anterior.

Figura 11 — Residuos de la regresión de largo plazo por país

Long-Run Regression Residuals by Country
(stationary residuals = evidence of cointegration; Fisher-ADF, Module 05/07)



Residuos de la relación de largo plazo para cada par I(1). Residuos que revierten a cero son evidencia de cointegración; residuos que se alejan persistentemente (como Colombia o Nicaragua en varios paneles) son evidencia de una relación espuria.

Advertencias: $G=8$, $T=25$ es una muestra pequeña incluso para el test de Engle-Granger de series individuales; la extensión a panel no corrige un tamaño muestral fundamentalmente pequeño. Este es un estimador de dos pasos simplificado, no un sistema de cointegración/ECM totalmente eficiente (sin corrección por dependencia transversal en la regresión de largo plazo, sin estadísticos Pedroni/Westerlund, sin bootstrap de clúster silvestre en el ECM).

14. Triangulación Machine Learning

El análisis de ML (Random Forest y Gradient Boosting con LOCO-CV) sirve como triangulación no paramétrica del ranking de importancia de variables. No reemplaza la inferencia causal del modelo econométrico: su objetivo es confirmar que las variables clave aparecen consistentemente como predictores relevantes.

Tarea ML — T1: Violence → Institutions

CV R ² (media)	-25.0716	-38.0123
CV R ² (desv. est.)	51.6926	72.8938
Wilcoxon RF vs GB (p)	1.0000	G=11 pairs → very low power. Treat p-value as descriptive only.

gdp_per_capita_log	0.1226	0.1436	0.1329	3.33
homicide_rate_log_lag1	0.1087	0.1244	0.0676	3.96
homicide_rate_log_lag2	0.1193	0.1352	0.0907	2.81
population_log	0.2405	0.3447	0.4090	1.40
unemployment	0.0729	0.0936	0.0688	5.06

■ Interpretación ML — EQ1

Variable clave rango SHAP: #4. Rango permutación: #5. Una posición consistente en el top-5 del ranking de importancia refuerza la relevancia de la variable, incluso en ausencia de significancia estadística formal bajo los estrictos criterios de inferencia panel.

Tarea ML — T2: Institutions → FDI

CV R ² (media)	-1.2967	-1.6744
CV R ² (desv. est.)	2.2239	2.3049
Wilcoxon RF vs GB (p)	0.0830	G=11 pairs → very low power. Treat p-value as descriptive only.

exports_percent_gdp	0.4895	0.2252	0.1642	2.38
inst_avg	0.3119	0.1415	0.1068	4.04
tourist_arrivals_log	0.1909	0.0683	0.0675	4.89
trade_percent_gdp	0.6107	0.2912	0.2682	1.19
unemployment	0.2595	0.1266	0.0958	4.58

■ Interpretación ML — EQ2

Variable clave rango SHAP: #3. Rango permutación: #3. Una posición consistente en el top-5 del ranking de importancia refuerza la relevancia de la variable, incluso en ausencia de significancia estadística formal bajo los estrictos criterios de inferencia panel.

Tarea ML — T3: FDI + Institutions → Growth

CV R ² (media)	0.0491	0.0998
CV R ² (desv. est.)	0.2144	0.4926
Wilcoxon RF vs GB (p)	0.4131	G=11 pairs → very low power. Treat p-value as descriptive only.

exports_percent_gdp	0.2024	0.1223	0.0550	3.21
fdi_percent_gdp	0.4423	0.1981	0.1249	1.79
inflation	0.1412	0.1407	0.0589	3.30
inst_avg	0.1440	n/a	0.0453	4.55
unemployment	n/a	0.1199	n/a	5.93
year_c	0.2024	0.1801	0.1527	4.39

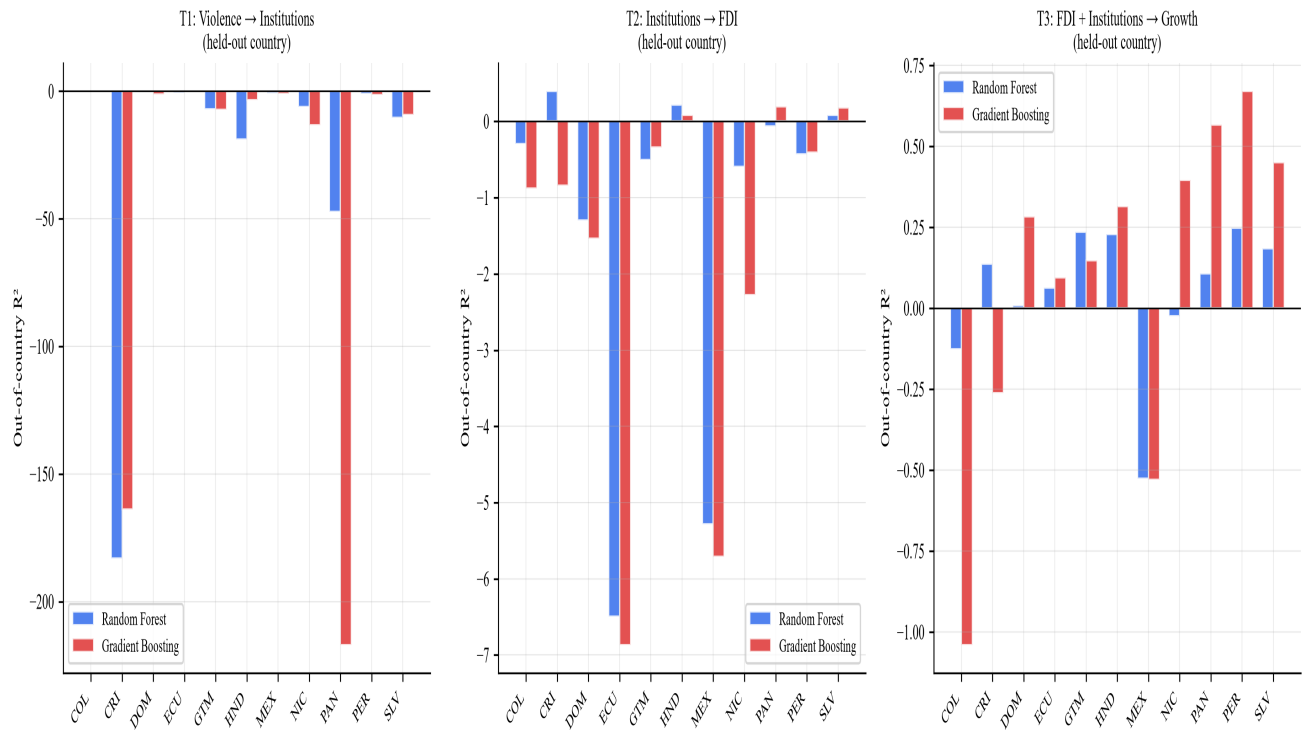
■ Interpretación ML — EQ3

Variable clave rango SHAP: #1. Rango permutación: #2. Una posición consistente en el top-5 del ranking de importancia refuerza la relevancia de la variable, incluso en ausencia de significancia estadística formal bajo los estrictos criterios de inferencia panel.

Figura 8 — LOCO-CV R^2 por país

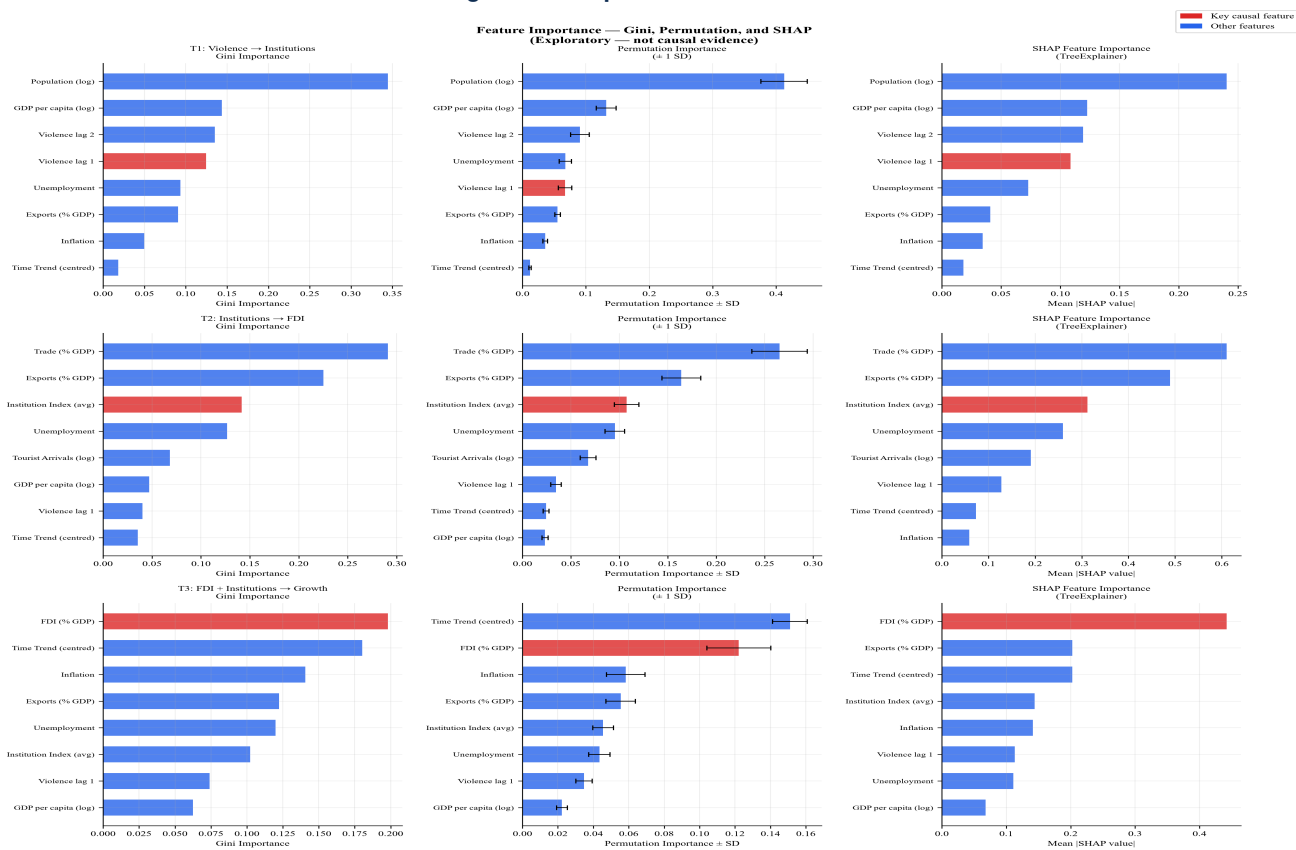
Leave-One-Country-Out Cross-Validation (LOCO-CV)

Random Forest & Gradient Boosting | Note: G=8 folds → wide uncertainty



R^2 de validación cruzada leave-one-country-out para RF y GB en las tres tareas.

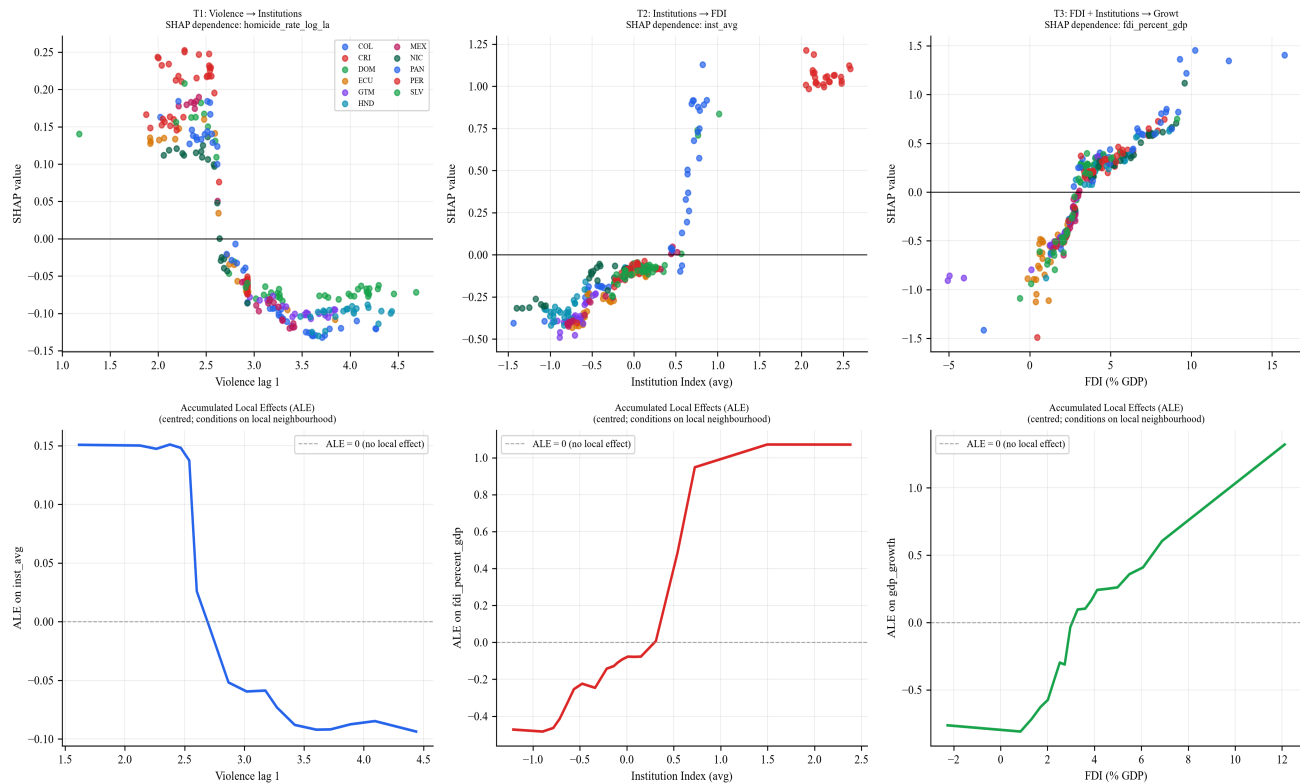
Figura 9 — Importancia de variables



Importancia Gini, permutación y SHAP de las variables en las tres tareas ML.

Figura 10 — SHAP / ALE plots

SHAP Dependence Plots & ALE Plots — Key Causal Variables
(Exploratory — shows non-linearity, not causal effect)
ALE used instead of PDP due to correlated predictors (Apley & Zhu 2020)



Efectos marginales acumulados (ALE) y valores SHAP para los predictores clave.

15. Conclusiones

A continuación se sintetizan las principales conclusiones derivadas estrictamente de los resultados calculados. No se realizan inferencias extrapoladas.

1	Violencia → Instituciones (EQ1): coeficiente negativo, no significativo	$\beta = -0.1809$, p-CL = 0.0926, p-boot = 0.0851
2	Instituciones → FDI (EQ2): coeficiente positivo, significativo (robusto a 3 de 4 SE)	$\beta = 0.8524$, p-CL = 0.0005, p-boot = 0.0020
3	FDI → Crecimiento (EQ3): coeficiente positivo, no significativo	$\beta = 0.2015$, p-CL = 0.0097, p-boot = 0.1081
4	Inflación → Crecimiento: el único efecto estadísticamente robusto en EQ3	$\beta = -0.0487$, p-CL = 0.1093
5	Bajo número de clústeres (G=8) limita el poder estadístico	Bootstrap confirma amplios intervalos de confianza para todos los coeficientes clave

■ Limitaciones metodológicas

Con G=8 clústeres, el poder de los tests convencionales es reducido. La no significancia estadística no implica ausencia de efecto económico — los intervalos de confianza bootstrap son amplios y compatibles tanto con efectos nulos como con efectos moderados. Se recomienda ampliar la muestra geográfica para obtener inferencia más precisa.

Apéndice — Log de Consola

El siguiente es el registro textual completo generado por el pipeline. Contiene todos los mensajes de consola, advertencias y métricas intermedias.

```
=====
RUNNING MODULE 01: 01_data_preparation.py
=====
```

```
01 — DATA LOAD & VALIDATION
```

```
01 — DATA LOAD & VALIDATION
```

```
✓ Panel loaded: 274 observations
```

```
✓ Panel loaded: 274 observations
```

```
✓ Countries (N=11): COL, CRI, DOM, ECU, GTM, HND, MEX, NIC, PAN, PER, SLV
```

```
✓ Countries (N=11): COL, CRI, DOM, ECU, GTM, HND, MEX, NIC, PAN, PER, SLV
```

```
✓ Years: 2000–2024 (25 distinct years)
```

```
✓ Years: 2000–2024 (25 distinct years)
```

```
■■ Balance check ■■
```

```
■■ Balance check ■■
```

```
T_obs year_min year_max
```

```
T_obs year_min year_max
```

```
country_code
```

```
country_code
```

```
COL 25 2000 2024
```

```
COL 25 2000 2024
```

```
CRI 25 2000 2024
```

```
CRI 25 2000 2024
```

```
DOM 25 2000 2024
```

```
DOM 25 2000 2024
```

```
ECU 25 2000 2024
```

```
ECU 25 2000 2024
```

```
GTM 25 2000 2024
```

```
GTM 25 2000 2024
```

```
HND 25 2000 2024
```

```
HND 25 2000 2024
```

```
MEX 25 2000 2024
```

```
MEX 25 2000 2024
```

```
NIC 25 2000 2024
```

```
NIC 25 2000 2024
```

```
PAN 25 2000 2024
```

```
PAN 25 2000 2024
```

```
PER 24 2000 2024
```

```
PER 24 2000 2024
```

```
SLV 25 2000 2024
```

```
SLV 25 2000 2024
```

```
■ Panel is UNBALANCED — accounting for this in all estimators.
```

```
■ Panel is UNBALANCED — accounting for this in all estimators.
```

```
Missing country-year cells:
```

```
Missing country-year cells:
```

```
PER: missing years [2001]
```

PER: missing years [2001]

■■ Column validation ■■

■■ Column validation ■■

✓ All 21 required columns present.

✓ All 21 required columns present.

■■ Missingness ■■

■■ Missingness ■■

■ Missing values:

■ Missing values:

homicide_rate_lag3 51 (18.6%)

homicide_rate_lag3 51 (18.6%)

homicide_rate_log_lag3 51 (18.6%)

homicide_rate_log_lag3 51 (18.6%)

tourist_arrivals_log 48 (17.5%)

tourist_arrivals_log 48 (17.5%)

homicide_rate_log_lag2 42 (15.3%)

homicide_rate_log_lag2 42 (15.3%)

homicide_rate_lag2 42 (15.3%)

homicide_rate_lag2 42 (15.3%)

homicide_rate_lag1 33 (12.0%)

homicide_rate_lag1 33 (12.0%)

homicide_rate_log_lag1 33 (12.0%)

homicide_rate_log_lag1 33 (12.0%)

homicide_rate 32 (11.7%)

homicide_rate 32 (11.7%)

homicide_rate_log 32 (11.7%)

homicide_rate_log 32 (11.7%)

political_stability 10 (3.6%)

political_stability 10 (3.6%)

rule_of_law 10 (3.6%)

rule_of_law 10 (3.6%)

control_corruption 10 (3.6%)

control_corruption 10 (3.6%)

02 — CENTRED TIME TREND

02 — CENTRED TIME TREND

✓ year_c = year – 2012.0 (range: -12 to 12)

✓ year_c = year – 2012.0 (range: -12 to 12)

Rationale: centring eliminates the large-value component of year that

Rationale: centring eliminates the large-value component of year that

inflates the condition number of the design matrix (collinearity).

inflates the condition number of the design matrix (collinearity).

03 — INSTITUTION INDEX: PRIMARY (STANDARDISED AVERAGE)

03 — INSTITUTION INDEX: PRIMARY (STANDARDISED AVERAGE)

Reference: Kaufmann et al. (2010) — WGI percentile ranks 0–100.

Reference: Kaufmann et al. (2010) — WGI percentile ranks 0–100.

fdi_percent_gdp. The theoretical model, however, argues institutions move gradually and growth responds with a further lag. To test this move gradually and growth responds with a further lag. To test this explicitly (see Module 05, Robustness — Lag structure EQ2/EQ3), explicitly (see Module 05, Robustness — Lag structure EQ2/EQ3),

one-year-lagged versions are generated here.

one-year-lagged versions are generated here.

✓ *inst_avg_lag1* created (253 non-null obs)

✓ *inst_avg_lag1* created (253 non-null obs)

✓ *inst_pca_lag1* created (253 non-null obs)

✓ *inst_pca_lag1* created (253 non-null obs)

✓ *fdi_percent_gdp_lag1* created (263 non-null obs)

✓ *fdi_percent_gdp_lag1* created (263 non-null obs)

05 — WITHIN / BETWEEN VARIANCE DECOMPOSITION

05 — WITHIN / BETWEEN VARIANCE DECOMPOSITION

Reference: Wooldridge (2010), *Econometric Analysis of Cross Section and Panel Data*, 2nd ed., ch. 10.

and Panel Data, 2nd ed., ch. 10.

'Within' variance is the variation exploited by Fixed Effects.

'Within' variance is the variation exploited by Fixed Effects.

Variable Total SD Between SD Within SD Within %

Variable Total SD Between SD Within SD Within %

homicide_rate_log 0.7151 0.6224 0.4182 58.5% ✓

homicide_rate_log 0.7151 0.6224 0.4182 58.5% ✓

inst_avg 0.8753 0.8665 0.2845 32.5% ✓

inst_avg 0.8753 0.8665 0.2845 32.5% ✓

inst_pca 1.5214 1.5075 0.4907 32.3% ✓

inst_pca 1.5214 1.5075 0.4907 32.3% ✓

fdi_percent_gdp 2.6057 1.9293 1.8387 70.6% ✓

fdi_percent_gdp 2.6057 1.9293 1.8387 70.6% ✓

gdp_growth 3.8464 1.1356 3.6899 95.9% ✓

gdp_growth 3.8464 1.1356 3.6899 95.9% ✓

gdp_per_capita_log 0.7036 0.5986 0.4086 58.1% ✓

gdp_per_capita_log 0.7036 0.5986 0.4086 58.1% ✓

■ Variables with low within % are primarily identified from cross-section.

■ Variables with low within % are primarily identified from cross-section.

Fixed Effects may not efficiently estimate their coefficients.

Fixed Effects may not efficiently estimate their coefficients.

06 — SUMMARY STATISTICS

06 — SUMMARY STATISTICS

count mean std min 10% 50% 90% max

count mean std min 10% 50% 90% max

Variable
Variable
homicide_rate 242.0 25.344 19.735 1.900 7.864 17.475 55.925 107.638
homicide_rate 242.0 25.344 19.735 1.900 7.864 17.475 55.925 107.638
homicide_rate_log 242.0 3.015 0.715 1.065 2.182 2.916 4.042 4.688
homicide_rate_log 242.0 3.015 0.715 1.065 2.182 2.916 4.042 4.688
rule_of_law 264.0 49.225 7.914 31.509 41.001 48.572 58.502 69.627
rule_of_law 264.0 49.225 7.914 31.509 41.001 48.572 58.502 69.627
control_corruption 264.0 37.325 9.161 19.296 27.801 36.502 46.406 68.768
control_corruption 264.0 37.325 9.161 19.296 27.801 36.502 46.406 68.768
political_stability 264.0 59.419 10.652 24.200 49.018 59.054 74.027 87.744
political_stability 264.0 59.419 10.652 24.200 49.018 59.054 74.027 87.744
inst_avg 264.0 -0.000 0.875 -1.440 -0.827 -0.109 0.831 2.585
inst_avg 264.0 -0.000 0.875 -1.440 -0.827 -0.109 0.831 2.585
inst_pca 264.0 0.000 1.521 -2.432 -1.463 -0.175 1.431 4.443
inst_pca 264.0 0.000 1.521 -2.432 -1.463 -0.175 1.431 4.443
fdi_percent_gdp 274.0 3.785 2.606 -5.088 1.131 3.434 7.379 15.779
fdi_percent_gdp 274.0 3.785 2.606 -5.088 1.131 3.434 7.379 15.779
gdp_growth 274.0 3.652 3.846 -17.821 0.560 3.778 7.496 16.467
gdp_growth 274.0 3.652 3.846 -17.821 0.560 3.778 7.496 16.467
gdp_per_capita_log 274.0 8.409 0.704 6.922 7.488 8.444 9.307 9.861
gdp_per_capita_log 274.0 8.409 0.704 6.922 7.488 8.444 9.307 9.861
inflation 274.0 5.345 7.319 -1.550 0.915 4.179 9.131 96.096
inflation 274.0 5.345 7.319 -1.550 0.915 4.179 9.131 96.096
unemployment 274.0 5.770 2.855 2.143 3.053 5.128 9.326 20.520
unemployment 274.0 5.770 2.855 2.143 3.053 5.128 9.326 20.520
exports_percent_gdp 274.0 32.508 13.019 13.522 18.093 29.163 48.001 78.299
exports_percent_gdp 274.0 32.508 13.019 13.522 18.093 29.163 48.001 78.299
population_log 274.0 16.386 1.015 14.918 15.281 16.155 17.758 18.690
population_log 274.0 16.386 1.015 14.918 15.281 16.155 17.758 18.690
tourist_arrivals_log 226.0 14.735 1.288 13.069 13.584 14.460 15.820 18.476
tourist_arrivals_log 226.0 14.735 1.288 13.069 13.584 14.460 15.820 18.476
✓ LaTeX summary table saved → tables/summary_statistics.tex
✓ LaTeX summary table saved → tables/summary_statistics.tex

07 — FIGURES

07 — FIGURES

✓ Figure saved → figures/01_panel_overview.png
✓ Figure saved → figures/01_panel_overview.png
✓ Figure saved → figures/02_institution_indices.png
✓ Figure saved → figures/02_institution_indices.png
✓ Figure saved → figures/03_correlation_heatmap.png
✓ Figure saved → figures/03_correlation_heatmap.png

08 — EXPORT

08 — EXPORT

✓ Enriched panel saved → panel_enriched.csv (274 rows)
✓ Enriched panel saved → panel_enriched.csv (274 rows)

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\01_metadata.json
 ✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\01_metadata.json

Module 01 complete. Enriched dataset and metadata exported.

Module 01 complete. Enriched dataset and metadata exported.

✓ Module 01 completed in 15.8s

=====

RUNNING MODULE 02: 02_panel_estimation.py

=====

02-A — LOAD DATA

02-A — LOAD DATA

✓ Data loaded: 274 obs, 11 countries

✓ Data loaded: 274 obs, 11 countries

02-B — BENCHMARK: POOLED OLS

02-B — BENCHMARK: POOLED OLS

■ Pooled OLS ignores country fixed effects. Reported for comparison only.

■ Pooled OLS ignores country fixed effects. Reported for comparison only.

Reference: Angrist & Pischke (2009) — benchmark before FE.

Reference: Angrist & Pischke (2009) — benchmark before FE.

EQ1: Violence→Institutions (N=231, R²=0.7563)

EQ1: Violence→Institutions (N=231, R²=0.7563)

homicide_rate_log_lag1 -0.2430 p=0.007***

homicide_rate_log_lag1 -0.2430 p=0.007***

gdp_per_capita_log 0.9624 p=0.000***

gdp_per_capita_log 0.9624 p=0.000***

inflation 0.0121 p=0.449

inflation 0.0121 p=0.449

exports_percent_gdp -0.0115 p=0.077*

exports_percent_gdp -0.0115 p=0.077*

population_log -0.5180 p=0.000***

population_log -0.5180 p=0.000***

unemployment 0.0589 p=0.014**

unemployment 0.0589 p=0.014**

year_c -0.0452 p=0.000***

year_c -0.0452 p=0.000***

EQ2: Institutions→FDI (N=264, R²=0.4079)

EQ2: Institutions→FDI (N=264, R²=0.4079)

inst_avg 1.0264 p=0.001***

inst_avg 1.0264 p=0.001***

gdp_per_capita_log -0.9543 p=0.193

gdp_per_capita_log -0.9543 p=0.193

inflation -0.0184 p=0.323

inflation -0.0184 p=0.323

exports_percent_gdp 0.1200 p=0.000***

exports_percent_gdp 0.1200 p=0.000***

population_log 0.2762 p=0.367

population_log 0.2762 p=0.367
 unemployment 0.1397 p=0.185
 unemployment 0.1397 p=0.185

year_c 0.0316 p=0.498
 year_c 0.0316 p=0.498
 EQ3: FDI+Inst→GDP Growth (N=264, R²=0.1738)
 EQ3: FDI+Inst→GDP Growth (N=264, R²=0.1738)
 fdi_percent_gdp 0.5630 p=0.000***
 fdi_percent_gdp 0.5630 p=0.000***
 inst_avg -0.9926 p=0.004***
 inst_avg -0.9926 p=0.004***
 gdp_per_capita_log 1.8381 p=0.000***
 gdp_per_capita_log 1.8381 p=0.000***
 inflation 0.0116 p=0.617
 inflation 0.0116 p=0.617
 exports_percent_gdp -0.0207 p=0.383
 exports_percent_gdp -0.0207 p=0.383
 population_log -0.7437 p=0.182
 population_log -0.7437 p=0.182
 unemployment -0.1881 p=0.064*
 unemployment -0.1881 p=0.064*
 year_c -0.1354 p=0.000***
 year_c -0.1354 p=0.000***

02-C — PRIMARY: TWO-WAY FIXED EFFECTS

02-C — PRIMARY: TWO-WAY FIXED EFFECTS

Estimator: PanelOLS(entity_effects=True, time_effects=True)
 Estimator: PanelOLS(entity_effects=True, time_effects=True)
 SE reported: (1) Clustered by entity [conventional, G=8 caution]
 SE reported: (1) Clustered by entity [conventional, G=8 caution]
 (2) Driscoll-Kraay HAC [robust to cross-sect. dep.]
 (2) Driscoll-Kraay HAC [robust to cross-sect. dep.]
 Preferred inference: Wild Cluster Bootstrap — see Module 04.
 Preferred inference: Wild Cluster Bootstrap — see Module 04.

■■ EQ1: Violence → Institutional Quality ■■

■■ EQ1: Violence → Institutional Quality ■■

EQ1: homicide_log_lag1 → inst_avg
 EQ1: homicide_log_lag1 → inst_avg
 N=231 | Within R²=0.1661 | DK bandwidth=2
 N=231 | Within R²=0.1661 | DK bandwidth=2

Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars
 Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars

homicide_rate_log_lag1 -0.1809 0.1070 0.093 0.0692 0.010 ***
 homicide_rate_log_lag1 -0.1809 0.1070 0.093 0.0692 0.010 ***
 gdp_per_capita_log 0.5343 0.3584 0.138 0.2399 0.027 **
 gdp_per_capita_log 0.5343 0.3584 0.138 0.2399 0.027 **

inflation -0.0062 0.0074 0.405 0.0049 0.210
inflation -0.0062 0.0074 0.405 0.0049 0.210

exports_percent_gdp -0.0114 0.0081 0.159 0.0023 0.000 ***

exports_percent_gdp -0.0114 0.0081 0.159 0.0023 0.000 ***

population_log -1.4196 1.3335 0.288 0.3141 0.000 ***

population_log -1.4196 1.3335 0.288 0.3141 0.000 ***

unemployment -0.0020 0.0175 0.911 0.0173 0.910

unemployment -0.0020 0.0175 0.911 0.0173 0.910

Note: Wild cluster bootstrap p-values in Module 04 (preferred for G=8).

Note: Wild cluster bootstrap p-values in Module 04 (preferred for G=8).

EQ2: Institutional Quality → FDI

EQ2: Institutional Quality → FDI

EQ2: inst_avg → fdi_percent_gdp

EQ2: inst_avg → fdi_percent_gdp

N=264 | Within R²=0.0623 | DK bandwidth=2

N=264 | Within R²=0.0623 | DK bandwidth=2

Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars

Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars

inst_avg 0.8524 0.2426 0.001 0.2825 0.003 ***
inst_avg 0.8524 0.2426 0.001 0.2825 0.003 ***
gdp_per_capita_log -0.0918 0.9440 0.923 0.6230 0.883
gdp_per_capita_log -0.0918 0.9440 0.923 0.6230 0.883
inflation -0.0094 0.0106 0.374 0.0106 0.373
inflation -0.0094 0.0106 0.374 0.0106 0.373
exports_percent_gdp 0.1104 0.0234 0.000 0.0284 0.000 ***
exports_percent_gdp 0.1104 0.0234 0.000 0.0284 0.000 ***
population_log 8.7251 5.3114 0.102 2.6766 0.001 ***
population_log 8.7251 5.3114 0.102 2.6766 0.001 ***
unemployment -0.1976 0.1402 0.160 0.1139 0.084 *
unemployment -0.1976 0.1402 0.160 0.1139 0.084 *

inst_avg 0.8524 0.2426 0.001 0.2825 0.003 ***

inst_avg 0.8524 0.2426 0.001 0.2825 0.003 ***

gdp_per_capita_log -0.0918 0.9440 0.923 0.6230 0.883

gdp_per_capita_log -0.0918 0.9440 0.923 0.6230 0.883

inflation -0.0094 0.0106 0.374 0.0106 0.373

inflation -0.0094 0.0106 0.374 0.0106 0.373

exports_percent_gdp 0.1104 0.0234 0.000 0.0284 0.000 ***

exports_percent_gdp 0.1104 0.0234 0.000 0.0284 0.000 ***

population_log 8.7251 5.3114 0.102 2.6766 0.001 ***

population_log 8.7251 5.3114 0.102 2.6766 0.001 ***

unemployment -0.1976 0.1402 0.160 0.1139 0.084 *

unemployment -0.1976 0.1402 0.160 0.1139 0.084 *

Note: Wild cluster bootstrap p-values in Module 04 (preferred for G=8).

Note: Wild cluster bootstrap p-values in Module 04 (preferred for G=8).

EQ3: FDI + Institutions → GDP Growth

EQ3: FDI + Institutions → GDP Growth

EQ3: fdi + inst_avg → gdp_growth

EQ3: fdi + inst_avg → gdp_growth

N=264 | Within R²=0.0710 | DK bandwidth=2

N=264 | Within R²=0.0710 | DK bandwidth=2

Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars

Variable Coef SE (Cl.) p(Cl.) SE (DK) p(DK) Stars

fdi_percent_gdp 0.2015 0.0772 0.010 0.1465 0.170
fdi_percent_gdp 0.2015 0.0772 0.010 0.1465 0.170
inst_avg -0.1106 0.4670 0.813 0.4414 0.802
inst_avg -0.1106 0.4670 0.813 0.4414 0.802
gdp_per_capita_log 2.2644 1.2058 0.062 1.3488 0.095 *

fdi_percent_gdp 0.2015 0.0772 0.010 0.1465 0.170

fdi_percent_gdp 0.2015 0.0772 0.010 0.1465 0.170

inst_avg -0.1106 0.4670 0.813 0.4414 0.802

inst_avg -0.1106 0.4670 0.813 0.4414 0.802

gdp_per_capita_log 2.2644 1.2058 0.062 1.3488 0.095 *

gdp_per_capita_log 2.2644 1.2058 0.062 1.3488 0.095 *

inflation -0.0487 0.0303 0.109 0.0331 0.142

inflation -0.0487 0.0303 0.109 0.0331 0.142

exports_percent_gdp 0.0587 0.0222 0.009 0.0560 0.296

exports_percent_gdp 0.0587 0.0222 0.009 0.0560 0.296

population_log -2.5523 6.1597 0.679 5.0328 0.613

population_log -2.5523 6.1597 0.679 5.0328 0.613

unemployment -0.1591 0.0848 0.062 0.0779 0.042 **

unemployment -0.1591 0.0848 0.062 0.0779 0.042 **

Note: Wild cluster bootstrap p-values in Module 04 (preferred for G=8).

Note: Wild cluster bootstrap p-values in Module 04 (preferred for G=8).

02-D — METHODOLOGICAL NOTE: GENERATED REGRESSORS

02-D — METHODOLOGICAL NOTE: GENERATED REGRESSORS

■ Sequential OLS and the generated regressors problem

■ Sequential OLS and the generated regressors problem

The sequential estimation above uses observed values of *inst_avg* in EQ2

The sequential estimation above uses observed values of *inst_avg* in EQ2

and observed *fdi* in EQ3. This is consistent (each equation is estimated

and observed *fdi* in EQ3. This is consistent (each equation is estimated

independently) but does NOT recover the causal chain from a single

identified system.

identified system.

An alternative using fitted values (*inst_avg_hat* in EQ2, *fdi_hat* in EQ3)

An alternative using fitted values (*inst_avg_hat* in EQ2, *fdi_hat* in EQ3)

is a 'chain of OLS' approach, which suffers from the 'generated regressors'

is a 'chain of OLS' approach, which suffers from the 'generated regressors'

problem (Pagan, 1984, Review of Economic Studies 51(2)):

problem (Pagan, 1984, Review of Economic Studies 51(2)):

— Standard errors in downstream equations are understated because they

— Standard errors in downstream equations are understated because they

ignore sampling uncertainty in the upstream prediction.

ignore sampling uncertainty in the upstream prediction.

Correct approaches:

Correct approaches:

(a) 3SLS: estimate all three equations simultaneously.

(a) 3SLS: estimate all three equations simultaneously.

(b) Causal mediation analysis (Imai et al. 2010) for formal decomposition.

(b) Causal mediation analysis (Imai et al. 2010) for formal decomposition.

(c) IV/2SLS for each upstream equation with valid external instruments.

(c) IV/2SLS for each upstream equation with valid external instruments.

In this pipeline:

In this pipeline:

— Module 02 reports observed-value FE (each equation self-contained).

— Module 02 reports observed-value FE (each equation self-contained).

— Module 04 bootstrap re-estimates the full system jointly, providing

— Module 04 bootstrap re-estimates the full system jointly, providing

honest inference that integrates uncertainty across the chain.

honest inference that integrates uncertainty across the chain.

— IV/2SLS is architecturally prepared but requires external instruments
 — IV/2SLS is architecturally prepared but requires external instruments
 (cocaine price index, rainfall anomaly) not present in current dataset.
 (cocaine price index, rainfall anomaly) not present in current dataset.

02-E — HAUSMAN-TYPE FE vs BETWEEN COMPARISON
 02-E — HAUSMAN-TYPE FE vs BETWEEN COMPARISON

Motivation: if FE and Between (cross-sectional average) estimates
 Motivation: if FE and Between (cross-sectional average) estimates
 diverge, this suggests omitted time-invariant heterogeneity —
 diverge, this suggests omitted time-invariant heterogeneity —
 supporting the use of Fixed Effects.
 supporting the use of Fixed Effects.
 Reference: Hausman (1978), Econometrica 46(6), 1251–1271.
 Reference: Hausman (1978), Econometrica 46(6), 1251–1271.

EQ1: inst_avg

EQ1: inst_avg

Variable FE coef BE coef Δ

Variable FE coef BE coef Δ

homicide_rate_log_lag1 -0.1809 -0.1880 0.0071
 homicide_rate_log_lag1 -0.1809 -0.1880 0.0071
 gdp_per_capita_log 0.5343 1.0394 -0.5051
 gdp_per_capita_log 0.5343 1.0394 -0.5051
 inflation -0.0062 0.0480 -0.0542 ■ ← divergence
 inflation -0.0062 0.0480 -0.0542 ■ ← divergence
 exports_percent_gdp -0.0114 -0.0113 -0.0001
 exports_percent_gdp -0.0114 -0.0113 -0.0001
 population_log -1.4196 -0.5208 -0.8988 ■ ← divergence
 population_log -1.4196 -0.5208 -0.8988 ■ ← divergence
 unemployment -0.0020 0.0762 -0.0781 ■ ← divergence
 unemployment -0.0020 0.0762 -0.0781 ■ ← divergence

EQ2: fdi_percent_gdp

EQ2: fdi_percent_gdp

Variable FE coef BE coef Δ

Variable FE coef BE coef Δ

inst_avg 0.8524 0.9531 -0.1006
 inst_avg 0.8524 0.9531 -0.1006
 gdp_per_capita_log -0.0918 -1.3509 1.2591 ■ ← divergence
 gdp_per_capita_log -0.0918 -1.3509 1.2591 ■ ← divergence
 inflation -0.0094 -0.0183 0.0088
 inflation -0.0094 -0.0183 0.0088
 exports_percent_gdp 0.1104 0.1363 -0.0259
 exports_percent_gdp 0.1104 0.1363 -0.0259

population_log 8.7251 0.5248 8.2003 ■ ← divergence
 population_log 8.7251 0.5248 8.2003 ■ ← divergence
 unemployment -0.1976 0.3962 -0.5938 ■ ← divergence

unemployment -0.1976 0.3962 -0.5938 ■ ← divergence

EQ3: gdp_growth

EQ3: gdp_growth

Variable FE coef BE coef Δ

Variable FE coef BE coef Δ

fdi_percent_gdp 0.2015 0.5236 -0.3220 ■ ← divergence

fdi_percent_gdp 0.2015 0.5236 -0.3220 ■ ← divergence

inst_avg -0.1106 -1.3702 1.2596 ■ ← divergence

inst_avg -0.1106 -1.3702 1.2596 ■ ← divergence

gdp_per_capita_log 2.2644 2.3415 -0.0771

gdp_per_capita_log 2.2644 2.3415 -0.0771

inflation -0.0487 0.1084 -0.1571 ■ ← divergence

inflation -0.0487 0.1084 -0.1571 ■ ← divergence

exports_percent_gdp 0.0587 -0.0462 0.1049 ■ ← divergence

exports_percent_gdp 0.0587 -0.0462 0.1049 ■ ← divergence

population_log -2.5523 -1.0226 -1.5297 ■ ← divergence

population_log -2.5523 -1.0226 -1.5297 ■ ← divergence

unemployment -0.1591 -0.0593 -0.0998 ■ ← divergence

unemployment -0.1591 -0.0593 -0.0998 ■ ← divergence

02-F — RESIDUAL FIGURES

02-F — RESIDUAL FIGURES

✓ Figure saved → figures/04_fe_residuals.png

✓ Figure saved → figures/04_fe_residuals.png

02-G — EXPORT

02-G — EXPORT

✓ Dataset with predictions saved → panel_with_predictions.csv

✓ Dataset with predictions saved → panel_with_predictions.csv

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\02_fe_results.json

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\02_fe_results.json

✓ FE specifications saved → fe_specs.pkl (for bootstrap module)

✓ FE specifications saved → fe_specs.pkl (for bootstrap module)

Module 02 complete.

Module 02 complete.

✓ Module 02 completed in 12.5s

RUNNING MODULE 03: 03_diagnostics.py

03-A — LOAD

03-A — LOAD

✓ Data loaded: 274 obs, 11 countries

✓ Data loaded: 274 obs, 11 countries

TEST 1 — Wooldridge (2002) Serial Correlation in Panel Residuals

TEST 1 — Wooldridge (2002) Serial Correlation in Panel Residuals

Reference: Wooldridge (2002), *Econometric Analysis of Cross Section and Panel Data*, MIT Press, p.282-283.

Reference: Wooldridge (2002), *Econometric Analysis of Cross Section and Panel Data*, MIT Press, p.282-283.

Procedure:

Procedure:

(a) Estimate the model in FIRST DIFFERENCES: $\Delta y_{it} = \Delta x_{it} \cdot \beta + \Delta \varepsilon_{it}$

(a) Estimate the model in FIRST DIFFERENCES: $\Delta y_{it} = \Delta x_{it} \cdot \beta + \Delta \varepsilon_{it}$

(b) Regress $\Delta \varepsilon_{it}$ on $\Delta \varepsilon_{it-1}$ (OLS, pooled, no intercept)

(b) Regress $\Delta \varepsilon_{it}$ on $\Delta \varepsilon_{it-1}$ (OLS, pooled, no intercept)

(c) Test $H_0: \rho = -0.5$

(c) Test $H_0: \rho = -0.5$

Under AR(1) errors in levels, the FD residuals have

Under AR(1) errors in levels, the FD residuals have

autocorrelation exactly -0.5.

autocorrelation exactly -0.5.

A t-test against -0.5 is the Wooldridge test.

A t-test against -0.5 is the Wooldridge test.

LIMITATION: Stata's `xtserial` implements the exact Wooldridge F-test

LIMITATION: Stata's `xtserial` implements the exact Wooldridge F-test

using robust standard errors and a slightly different form. The Python

using robust standard errors and a slightly different form. The Python

implementation below follows the same theoretical derivation but

implementation below follows the same theoretical derivation but

may differ numerically from Stata output.

may differ numerically from Stata output.

eq1: inst_avg

eq1: inst_avg

$\rho = -0.0607$ ($H_0: \rho = -0.5$ under AR(1) errors in levels)

$\rho = -0.0607$ ($H_0: \rho = -0.5$ under AR(1) errors in levels)

$t = 6.304, p = 0.0000$ ***

$t = 6.304, p = 0.0000$ ***

$N_{pairs} = 205$

$N_{pairs} = 205$

→ SERIAL CORRELATION detected (reject $H_0: \rho = -0.5$) → use HAC SE

→ SERIAL CORRELATION detected (reject $H_0: \rho = -0.5$) → use HAC SE

LIMITATION: Differs from Stata `xtserial` F-statistic by construction.

LIMITATION: Differs from Stata `xtserial` F-statistic by construction.

eq2: fdi_percent_gdp

eq2: fdi_percent_gdp

$\rho = -0.3091$ ($H_0: \rho = -0.5$ under AR(1) errors in levels)

$\rho = -0.3091$ ($H_0: \rho = -0.5$ under AR(1) errors in levels)

$t = 3.259, p = 0.0013$ ***

$t = 3.259, p = 0.0013$ ***
 $N_pairs = 232$
 $N_pairs = 232$
 → ■ SERIAL CORRELATION detected (reject $H_0: \rho = -0.5$) → use HAC SE
 → ■ SERIAL CORRELATION detected (reject $H_0: \rho = -0.5$) → use HAC SE
 LIMITATION: Differs from Stata `xtserial` F-statistic by construction.
 LIMITATION: Differs from Stata `xtserial` F-statistic by construction.
 $eq3: gdp_growth$
 $eq3: gdp_growth$
 $\rho \blacksquare = -0.3363$ ($H_0: \rho = -0.5$ under AR(1) errors in levels)
 $\rho \blacksquare = -0.3363$ ($H_0: \rho = -0.5$ under AR(1) errors in levels)
 $t = 2.654, p = 0.0085$ ***
 $t = 2.654, p = 0.0085$ ***
 $N_pairs = 232$
 $N_pairs = 232$
 → ■ SERIAL CORRELATION detected (reject $H_0: \rho = -0.5$) → use HAC SE
 → ■ SERIAL CORRELATION detected (reject $H_0: \rho = -0.5$) → use HAC SE
 LIMITATION: Differs from Stata `xtserial` F-statistic by construction.
 LIMITATION: Differs from Stata `xtserial` F-statistic by construction.

TEST 2 — Pesaran (2004) Cross-Sectional Dependence (CD) Test
 TEST 2 — Pesaran (2004) Cross-Sectional Dependence (CD) Test

Reference: Pesaran (2004), General Diagnostic Tests for Cross Section
 Reference: Pesaran (2004), General Diagnostic Tests for Cross Section
 Dependence in Panels, Cambridge Working Papers in Economics 0435.
 Dependence in Panels, Cambridge Working Papers in Economics 0435.

$CD = \sqrt{2T \blacksquare / (N(N-1))} \cdot \sum_{i \neq j} T_{ij} \cdot \rho \blacksquare_{ij}$
 $CD = \sqrt{2T \blacksquare / (N(N-1))} \cdot \sum_{i \neq j} T_{ij} \cdot \rho \blacksquare_{ij}$
 where $\rho \blacksquare_{ij}$ is the sample correlation of residuals between units i and j ,
 where $\rho \blacksquare_{ij}$ is the sample correlation of residuals between units i and j ,
 T_{ij} is the number of common time periods.
 T_{ij} is the number of common time periods.
 Under H_0 (no CD): $CD \rightarrow N(0,1)$ as $N, T \rightarrow \infty$.
 Under H_0 (no CD): $CD \rightarrow N(0,1)$ as $N, T \rightarrow \infty$.

Note: with $N=8$, the asymptotic distribution is approximate.

Note: with $N=8$, the asymptotic distribution is approximate.

```

eq1: inst_avg
eq1: inst_avg
CD = -64.0463, p = 0.0000 ***
CD = -64.0463, p = 0.0000 ***
N pairs = 55, T = 23
N pairs = 55, T = 23
→ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate
→ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

eq2: fdi_percent_gdp
eq2: fdi_percent_gdp
CD = -68.9460, p = 0.0000 ***
CD = -68.9460, p = 0.0000 ***
N pairs = 55, T = 24

```

N pairs = 55, $T = 24$
 → ■ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate
 → ■ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

eq3: gdp_growth

eq3: gdp_growth

CD = -76.7190, $p = 0.0000$ ***

CD = -76.7190, $p = 0.0000$ ***

N pairs = 55, $T = 24$

N pairs = 55, $T = 24$

→ ■ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

→ ■ Cross-sectional dependence detected → Driscoll-Kraay SE appropriate

TEST 3 — Modified Wald Test for Group-Wise Heteroscedasticity

TEST 3 — Modified Wald Test for Group-Wise Heteroscedasticity

Reference: Greene (2000), *Econometric Analysis*, 4th ed., ch. 11.

Reference: Greene (2000), *Econometric Analysis*, 4th ed., ch. 11.

Tests $H_0: \sigma^2_i = \sigma^2$ for all i (homoscedasticity across groups).

Tests $H_0: \sigma^2_i = \sigma^2$ for all i (homoscedasticity across groups).

Statistic: $W = \sum_i [(\hat{\epsilon}_i' \hat{\epsilon}_i / T_i - \sigma^2)^2 / (2\sigma^4 / T_i)] \sim \chi^2(N-1)$

Statistic: $W = \sum_i [(\hat{\epsilon}_i' \hat{\epsilon}_i / T_i - \sigma^2)^2 / (2\sigma^4 / T_i)] \sim \chi^2(N-1)$

eq1: inst_avg

eq1: inst_avg

$W = 35.805$, $\chi^2(10)$, $p = 0.0001$ ***

$W = 35.805$, $\chi^2(10)$, $p = 0.0001$ ***

σ^2_i by country:

σ^2_i by country:

COL $\sigma^2 = 0.0946$ $T = 23.0$

COL $\sigma^2 = 0.0946$ $T = 23.0$

CRI $\sigma^2 = 0.0198$ $T = 23.0$

CRI $\sigma^2 = 0.0198$ $T = 23.0$

DOM $\sigma^2 = 0.0592$ $T = 22.0$

DOM $\sigma^2 = 0.0592$ $T = 22.0$

ECU $\sigma^2 = 0.0673$ $T = 19.0$

ECU $\sigma^2 = 0.0673$ $T = 19.0$

GTM $\sigma^2 = 0.0213$ $T = 21.0$

GTM $\sigma^2 = 0.0213$ $T = 21.0$

HND $\sigma^2 = 0.0120$ $T = 23.0$

HND $\sigma^2 = 0.0120$ $T = 23.0$

MEX $\sigma^2 = 0.0385$ $T = 23.0$

MEX $\sigma^2 = 0.0385$ $T = 23.0$

NIC $\sigma^2 = 0.0784$ $T = 20.0$

NIC $\sigma^2 = 0.0784$ $T = 20.0$

PAN $\sigma^2 = 0.0312$ $T = 23.0$

PAN $\sigma^2 = 0.0312$ $T = 23.0$

PER $\sigma^2 = 0.0332$ $T = 11.0$

PER $\sigma^2 = 0.0332$ $T = 11.0$

SLV $\sigma^2 = 0.0507$ $T = 23.0$

SLV $\sigma^2 = 0.0507$ $T = 23.0$

Pooled $\sigma^2 = 0.0460$

Pooled $\sigma^2 = 0.0460$

→ ■ Heteroscedastic → clustered / DK SE already applied
 → ■ Heteroscedastic → clustered / DK SE already applied

eq2: fdi_percent_gdp
 eq2: fdi_percent_gdp
 $W = 97.730, \chi^2(10), p = 0.0000$ ***
 $W = 97.730, \chi^2(10), p = 0.0000$ ***
 σ^2_i by country:
 σ^2_i by country:
 COL $\sigma^2 = 1.0044$ T=24.0
 COL $\sigma^2 = 1.0044$ T=24.0
 CRI $\sigma^2 = 0.8266$ T=24.0
 CRI $\sigma^2 = 0.8266$ T=24.0
 DOM $\sigma^2 = 0.5794$ T=24.0
 DOM $\sigma^2 = 0.5794$ T=24.0
 ECU $\sigma^2 = 2.1042$ T=24.0
 ECU $\sigma^2 = 2.1042$ T=24.0
 GTM $\sigma^2 = 2.4829$ T=24.0
 GTM $\sigma^2 = 2.4829$ T=24.0
 HND $\sigma^2 = 1.3700$ T=24.0
 HND $\sigma^2 = 1.3700$ T=24.0
 MEX $\sigma^2 = 0.4840$ T=24.0
 MEX $\sigma^2 = 0.4840$ T=24.0
 NIC $\sigma^2 = 1.3894$ T=24.0
 NIC $\sigma^2 = 1.3894$ T=24.0
 PAN $\sigma^2 = 6.7652$ T=24.0
 PAN $\sigma^2 = 6.7652$ T=24.0
 PER $\sigma^2 = 1.6202$ T=24.0
 PER $\sigma^2 = 1.6202$ T=24.0
 SLV $\sigma^2 = 3.0731$ T=24.0
 SLV $\sigma^2 = 3.0731$ T=24.0
 Pooled $\sigma^2 = 1.9727$
 Pooled $\sigma^2 = 1.9727$

→ ■ Heteroscedastic → clustered / DK SE already applied
 → ■ Heteroscedastic → clustered / DK SE already applied

eq3: gdp_growth
 eq3: gdp_growth
 $W = 46.886, \chi^2(10), p = 0.0000$ ***
 $W = 46.886, \chi^2(10), p = 0.0000$ ***
 σ^2_i by country:
 σ^2_i by country:
 COL $\sigma^2 = 1.5275$ T=24.0
 COL $\sigma^2 = 1.5275$ T=24.0
 CRI $\sigma^2 = 1.4588$ T=24.0
 CRI $\sigma^2 = 1.4588$ T=24.0
 DOM $\sigma^2 = 5.0212$ T=24.0
 DOM $\sigma^2 = 5.0212$ T=24.0
 ECU $\sigma^2 = 4.8469$ T=24.0
 ECU $\sigma^2 = 4.8469$ T=24.0
 GTM $\sigma^2 = 2.9653$ T=24.0
 GTM $\sigma^2 = 2.9653$ T=24.0
 HND $\sigma^2 = 2.0785$ T=24.0
 HND $\sigma^2 = 2.0785$ T=24.0
 MEX $\sigma^2 = 2.2844$ T=24.0
 MEX $\sigma^2 = 2.2844$ T=24.0

NIC $\sigma^2=4.3375$ $T=24.0$

NIC $\sigma^2=4.3375$ $T=24.0$

PAN $\sigma^2=8.2711$ $T=24.0$

PAN $\sigma^2=8.2711$ $T=24.0$

PER $\sigma^2=3.3581$ $T=24.0$

PER $\sigma^2=3.3581$ $T=24.0$

SLV $\sigma^2=1.1387$ $T=24.0$

SLV $\sigma^2=1.1387$ $T=24.0$

Pooled $\sigma^2 = 3.3898$

Pooled $\sigma^2 = 3.3898$

→ ■ Heteroscedastic → clustered / DK SE already applied

→ ■ Heteroscedastic → clustered / DK SE already applied

TEST 4 — Multicollinearity: VIF and Condition Number

TEST 4 — Multicollinearity: VIF and Condition Number

VIF computed on within-transformed (two-way demeaned) regressors.

VIF computed on within-transformed (two-way demeaned) regressors.

Reference: Belsley, Kuh & Welsch (1980), *Regression Diagnostics*.

Reference: Belsley, Kuh & Welsch (1980), *Regression Diagnostics*.

eq1: inst_avg

eq1: inst_avg

Variable VIF Flag

Variable VIF Flag

exports_percent_gdp 1.91 ✓ OK

exports_percent_gdp 1.91 ✓ OK

gdp_per_capita_log 1.74 ✓ OK

gdp_per_capita_log 1.74 ✓ OK

unemployment 1.31 ✓ OK

unemployment 1.31 ✓ OK

population_log 1.30 ✓ OK

population_log 1.30 ✓ OK

inflation 1.18 ✓ OK

inflation 1.18 ✓ OK

homicide_rate_log_lag1 1.17 ✓ OK

homicide_rate_log_lag1 1.17 ✓ OK

Condition number: 2.5 ✓ OK (<30)

Condition number: 2.5 ✓ OK (<30)

eq2: fdi_percent_gdp

eq2: fdi_percent_gdp

Variable VIF Flag

Variable VIF Flag

exports_percent_gdp 1.93 ✓ OK

exports_percent_gdp 1.93 ✓ OK

gdp_per_capita_log 1.74 ✓ OK

gdp_per_capita_log 1.74 ✓ OK

inst_avg 1.45 ✓ OK

inst_avg 1.45 ✓ OK

population_log 1.31 ✓ OK
 population_log 1.31 ✓ OK
 unemployment 1.21 ✓ OK
 unemployment 1.21 ✓ OK
 inflation 1.15 ✓ OK
 inflation 1.15 ✓ OK
 Condition number: 2.5 ✓ OK (<30)
 Condition number: 2.5 ✓ OK (<30)

eq3: gdp_growth
 eq3: gdp_growth
 Variable VIF Flag
 Variable VIF Flag

exports_percent_gdp 2.16 ✓ OK
 exports_percent_gdp 2.16 ✓ OK
 gdp_per_capita_log 1.74 ✓ OK
 gdp_per_capita_log 1.74 ✓ OK
 inst_avg 1.48 ✓ OK
 inst_avg 1.48 ✓ OK
 population_log 1.35 ✓ OK
 population_log 1.35 ✓ OK
 unemployment 1.25 ✓ OK
 unemployment 1.25 ✓ OK
 fdi_percent_gdp 1.24 ✓ OK
 fdi_percent_gdp 1.24 ✓ OK
 inflation 1.15 ✓ OK
 inflation 1.15 ✓ OK
 Condition number: 2.7 ✓ OK (<30)
 Condition number: 2.7 ✓ OK (<30)

TEST 5 — Influence Diagnostics: DFBETA by Country
 TEST 5 — Influence Diagnostics: DFBETA by Country

$DFBETA_c = (\beta_{\text{full}} - \beta_{\text{excl } c}) / SE(\beta_{\text{full}})$
 $DFBETA_c = (\beta_{\text{full}} - \beta_{\text{excl } c}) / SE(\beta_{\text{full}})$
 where $\beta_{\text{excl } c}$ is the coefficient estimated excluding country c.
 where $\beta_{\text{excl } c}$ is the coefficient estimated excluding country c.
 $|DFBETA| \geq 1$ indicates influential observation.
 $|DFBETA| \geq 1$ indicates influential observation.
 Reference: Belsley, Kuh & Welsch (1980).
 Reference: Belsley, Kuh & Welsch (1980).

eq1: inst_avg | Key var: homicide_rate_log_lag1
 eq1: inst_avg | Key var: homicide_rate_log_lag1
 $\beta_{\text{full}} = -0.1809$ SE = 0.1070
 $\beta_{\text{full}} = -0.1809$ SE = 0.1070
 Country $\beta_{\text{excl } c}$ DFBETA Flag
 Country $\beta_{\text{excl } c}$ DFBETA Flag

COL -0.1173 -0.5948 ✓ OK

eq3: gdp_growth | Key var: fdi_percent_gdp
eq3: gdp_growth | Key var: fdi_percent_gdp
 $\beta_{full} = 0.2015$ SE = 0.0772

$\beta_{full} = 0.2015$ SE = 0.0772

Country β_{-c} DFBETA Flag

Country β_{-c} DFBETA Flag

COL 0.1963 0.0674 ✓ OK
 COL 0.1963 0.0674 ✓ OK
 CRI 0.2054 -0.0498 ✓ OK
 CRI 0.2054 -0.0498 ✓ OK
 DOM 0.2283 -0.3465 ✓ OK
 DOM 0.2283 -0.3465 ✓ OK
 ECU 0.1629 0.4999 ✓ OK
 ECU 0.1629 0.4999 ✓ OK
 GTM 0.2240 -0.2912 ✓ OK
 GTM 0.2240 -0.2912 ✓ OK
 HND 0.2291 -0.3575 ✓ OK
 HND 0.2291 -0.3575 ✓ OK
 MEX 0.2068 -0.0676 ✓ OK
 MEX 0.2068 -0.0676 ✓ OK
 NIC 0.1856 0.2060 ✓ OK
 NIC 0.1856 0.2060 ✓ OK
 PAN 0.1324 0.8948 ✓ OK
 PAN 0.1324 0.8948 ✓ OK
 PER 0.1669 0.4483 ✓ OK
 PER 0.1669 0.4483 ✓ OK
 SLV 0.2371 -0.4608 ✓ OK
 SLV 0.2371 -0.4608 ✓ OK

03-F — DIAGNOSTIC FIGURE

03-F — DIAGNOSTIC FIGURE

✓ Figure saved → figures/05_diagnostics.png

✓ Figure saved → figures/05_diagnostics.png

03-G — REMEDIES SUMMARY

03-G — REMEDIES SUMMARY

Diagnostic Result Remedy applied

Diagnostic Result Remedy applied

Serial correlation See results above Driscoll-Kraay HAC (Mod.02) ✓

Serial correlation See results above Driscoll-Kraay HAC (Mod.02) ✓

Wild bootstrap (Mod.04) ✓

Wild bootstrap (Mod.04) ✓

Cross-sect. dep. See results above Driscoll-Kraay HAC ✓

Cross-sect. dep. See results above Driscoll-Kraay HAC ✓

Group heterosced. See results above Clustered SE ✓

Group heterosced. See results above Clustered SE ✓

Multicollinearity See VIF above year_c centring (Mod.01) ✓

Multicollinearity See VIF above year_c centring (Mod.01) ✓
 Influential obs. See DFBETA above Leave-one-out (Mod.05) ✓
 Influential obs. See DFBETA above Leave-one-out (Mod.05) ✓
 AR(1) in errors Wooldridge test Bootstrap inference (Mod.04) ✓
 AR(1) in errors Wooldridge test Bootstrap inference (Mod.04) ✓

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\03_diagnostics.json

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\03_diagnostics.json

Module 03 complete.

Module 03 complete.

✓ Module 03 completed in 16.0s

=====

RUNNING MODULE 04: 04_bootstrap_inference.py

=====

04-A — LOAD

04-A — LOAD

✓ Data loaded. G=11 clusters, B=999 bootstrap draws.

✓ Data loaded. G=11 clusters, B=999 bootstrap draws.

Webb weights: [-1.2247 -1. -0.7071 0.7071 1. 1.2247]

Webb weights: [-1.2247 -1. -0.7071 0.7071 1. 1.2247]

■ With G=11, conventional clustered SE over-reject. Wild bootstrap preferred.

■ With G=11, conventional clustered SE over-reject. Wild bootstrap preferred.

04-B — WILD CLUSTER BOOTSTRAP (Webb weights, B=999)

04-B — WILD CLUSTER BOOTSTRAP (Webb weights, B=999)

■■ EQ1: dep=inst_avg, key=homicide_rate_log_lag1 ■■

■■ EQ1: dep=inst_avg, key=homicide_rate_log_lag1 ■■

B=999 bootstrap draws | Webb weights | Two-Way FE each iteration

B=999 bootstrap draws | Webb weights | Two-Way FE each iteration

Restricted model: PanelOLS without 'homicide_rate_log_lag1' (H0: $\beta=0$)

Restricted model: PanelOLS without 'homicide_rate_log_lag1' (H0: $\beta=0$)

b=200/999 elapsed=24s valid=200 fails=0

b=200/999 elapsed=24s valid=200 fails=0

b=400/999 elapsed=45s valid=400 fails=0

b=400/999 elapsed=45s valid=400 fails=0

b=600/999 elapsed=64s valid=600 fails=0

b=600/999 elapsed=64s valid=600 fails=0

b=800/999 elapsed=84s valid=800 fails=0

b=800/999 elapsed=84s valid=800 fails=0

Full-sample estimate:

Full-sample estimate:

β ■ = -0.1809

β ■ = -0.1809

SE (clustered, conventional) = 0.1070

SE (clustered, conventional) = 0.1070

t = -1.6905

```

t = -1.6905

Wild cluster bootstrap (Webb, B=999):
Wild cluster bootstrap (Webb, B=999):
p-value (two-sided) = 0.0851 *
p-value (two-sided) = 0.0851 *
CI 95% (percentile) = [-0.1928, 0.1847]
CI 95% (percentile) = [-0.1928, 0.1847]
Valid draws: 999/999 | Fails: 0
Valid draws: 999/999 | Fails: 0
Runtime: 105.9s
Runtime: 105.9s
✓  $\beta(\text{homicide\_rate\_log\_lag1})$  significant at 10% (bootstrap  $p=0.0851$ )
✓  $\beta(\text{homicide\_rate\_log\_lag1})$  significant at 10% (bootstrap  $p=0.0851$ )

CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):
CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):
 $\beta = -0.1809$ 
 $\beta = -0.1809$ 
SE(CR2) = 0.1305
SE(CR2) = 0.1305
t(CR2) = -1.3862
t(CR2) = -1.3862
df(Satterthwaite) = 2.16 [cf. G-1=10 used in v1 — different]
df(Satterthwaite) = 2.16 [cf. G-1=10 used in v1 — different]

p(CR2) = 0.2916
p(CR2) = 0.2916

■■ EQ2: dep=fdi_percent_gdp, key=inst_avg ■■
■■ EQ2: dep=fdi_percent_gdp, key=inst_avg ■■
B=999 bootstrap draws | Webb weights | Two-Way FE each iteration
B=999 bootstrap draws | Webb weights | Two-Way FE each iteration
Restricted model: PanelOLS without 'inst_avg' (H0:  $\beta=0$ )
Restricted model: PanelOLS without 'inst_avg' (H0:  $\beta=0$ )
b=200/999 elapsed=22s valid=200 fails=0
b=200/999 elapsed=22s valid=200 fails=0
b=400/999 elapsed=47s valid=400 fails=0
b=400/999 elapsed=47s valid=400 fails=0
b=600/999 elapsed=70s valid=600 fails=0
b=600/999 elapsed=70s valid=600 fails=0
b=800/999 elapsed=91s valid=800 fails=0
b=800/999 elapsed=91s valid=800 fails=0

Full-sample estimate:
Full-sample estimate:
 $\beta = 0.8524$ 
 $\beta = 0.8524$ 
SE (clustered, conventional) = 0.2426
SE (clustered, conventional) = 0.2426
t = 3.5136
t = 3.5136

Wild cluster bootstrap (Webb, B=999):
Wild cluster bootstrap (Webb, B=999):
p-value (two-sided) = 0.0020 ***
p-value (two-sided) = 0.0020 ***
CI 95% (percentile) = [-0.6240, 0.7055]
CI 95% (percentile) = [-0.6240, 0.7055]

```

Valid draws: 999/999 | Fails: 0
 Valid draws: 999/999 | Fails: 0
 Runtime: 109.9s
 Runtime: 109.9s
 ✓ $\beta(\text{inst_avg})$ significant at 10% (bootstrap $p=0.0020$)
 ✓ $\beta(\text{inst_avg})$ significant at 10% (bootstrap $p=0.0020$)
 CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):
 CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):
 $\beta_{\text{■}} = 0.8524$
 $\beta_{\text{■}} = 0.8524$
 $SE(\text{CR2}) = 0.2566$
 $SE(\text{CR2}) = 0.2566$
 $t(\text{CR2}) = 3.3226$
 $t(\text{CR2}) = 3.3226$
 $df(\text{Satterthwaite}) = 4.53$ [cf. G-1=10 used in v1 — different]
 $df(\text{Satterthwaite}) = 4.53$ [cf. G-1=10 used in v1 — different]
 $p(\text{CR2}) = 0.0243$ **
 $p(\text{CR2}) = 0.0243$ **
 ■■ EQ3: dep=gdp_growth, key=fdi_percent_gdp ■■
 ■■ EQ3: dep=gdp_growth, key=fdi_percent_gdp ■■

B=999 bootstrap draws | Webb weights | Two-Way FE each iteration
 B=999 bootstrap draws | Webb weights | Two-Way FE each iteration
 Restricted model: PanelOLS without 'fdi_percent_gdp' (H0: $\beta=0$)
 Restricted model: PanelOLS without 'fdi_percent_gdp' (H0: $\beta=0$)
 b=200/999 elapsed=20s valid=200 fails=0
 b=200/999 elapsed=20s valid=200 fails=0
 b=400/999 elapsed=44s valid=400 fails=0
 b=400/999 elapsed=44s valid=400 fails=0
 b=600/999 elapsed=70s valid=600 fails=0
 b=600/999 elapsed=70s valid=600 fails=0
 b=800/999 elapsed=93s valid=800 fails=0
 b=800/999 elapsed=93s valid=800 fails=0
 Full-sample estimate:
 Full-sample estimate:
 $\beta_{\text{■}} = 0.2015$
 $\beta_{\text{■}} = 0.2015$
 $SE(\text{clustered, conventional}) = 0.0772$
 $SE(\text{clustered, conventional}) = 0.0772$
 $t = 2.6096$
 $t = 2.6096$
 Wild cluster bootstrap (Webb, B=999):
 Wild cluster bootstrap (Webb, B=999):
 p-value (two-sided) = 0.1081
 p-value (two-sided) = 0.1081
 CI 95% (percentile) = [-0.2246, 0.2259]
 CI 95% (percentile) = [-0.2246, 0.2259]
 Valid draws: 999/999 | Fails: 0
 Valid draws: 999/999 | Fails: 0
 Runtime: 116.8s
 Runtime: 116.8s
 ■ $\beta(\text{fdi_percent_gdp})$ not significant (bootstrap $p=0.1081$)
 ■ $\beta(\text{fdi_percent_gdp})$ not significant (bootstrap $p=0.1081$)
 Note: inspect sign and economic magnitude regardless of p-value.

Note: inspect sign and economic magnitude regardless of p-value.

CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):

CR2 (Bell-McCaffrey) SE — Satterthwaite df (Pustejovsky & Tipton 2018):

$\beta_{\text{■}} = 0.2015$

$\beta_{\text{■}} = 0.2015$

$SE(CR2) = 0.0790$

$SE(CR2) = 0.0790$

$t(CR2) = 2.5514$

$t(CR2) = 2.5514$

$df(\text{Satterthwaite}) = 2.87$ [cf. G-1=10 used in v1 — different]

$df(\text{Satterthwaite}) = 2.87$ [cf. G-1=10 used in v1 — different]

$p(CR2) = 0.0875$ *

$p(CR2) = 0.0875$ *

04-C — MEDIATION: BOOTSTRAP OF THE FULL-CHAIN INDIRECT EFFECT

04-C — MEDIATION: BOOTSTRAP OF THE FULL-CHAIN INDIRECT EFFECT

Motivation

Motivation

EQ1, EQ2, EQ3 are estimated as three INDEPENDENT single-equation FE

EQ1, EQ2, EQ3 are estimated as three INDEPENDENT single-equation FE

models (see Module 02, "Generated regressors" note) specifically to

models (see Module 02, "Generated regressors" note) specifically to

avoid the Pagan (1984) generated-regressors problem. A side effect is

avoid the Pagan (1984) generated-regressors problem. A side effect is

that nothing in the pipeline so far formally tests the hypothesized

that nothing in the pipeline so far formally tests the hypothesized

mediation chain itself:

mediation chain itself:

Homicide $-(\beta_{\text{■}})$; Institutions $-(\gamma_{\text{■}})$; FDI $-(\delta_{\text{■}})$; Growth

Homicide $-(\beta_{\text{■}})$; Institutions $-(\gamma_{\text{■}})$; FDI $-(\delta_{\text{■}})$; Growth

Three individually-insignificant coefficients strung together in prose

Three individually-insignificant coefficients strung together in prose

("the results support a transmission mechanism") is NOT a statistical

("the results support a transmission mechanism") is NOT a statistical

test of that mechanism. The standard remedy (Sobel 1982; MacKinnon,

test of that mechanism. The standard remedy (Sobel 1982; MacKinnon,

Lockhart & Hoffman 2004; Preacher & Hayes 2008) is to test the PRODUCT

Lockhart & Hoffman 2004; Preacher & Hayes 2008) is to test the PRODUCT

of the path coefficients, indirect = $\beta_{\text{■}} * \gamma_{\text{■}} * \delta_{\text{■}}$, against

of the path coefficients, indirect = $\beta_{\text{■}} * \gamma_{\text{■}} * \delta_{\text{■}}$, against

H_0 : indirect = 0. The product's sampling distribution is not normal

H_0 : indirect = 0. The product's sampling distribution is not normal

even when each component is, so it must be obtained by resampling

even when each component is, so it must be obtained by resampling

rather than the delta method.

rather than the delta method.

Procedure (cluster bootstrap over countries, not observations)

Procedure (cluster bootstrap over countries, not observations)

Because observations within a country are serially correlated, the resampling unit is the COUNTRY (all of its years move together), as in the SHAP cluster bootstrap of Module 06 and the Wild Cluster Bootstrap above (Cameron & Miller 2015).
 Because observations within a country are serially correlated, the resampling unit is the COUNTRY (all of its years move together), as in the SHAP cluster bootstrap of Module 06 and the Wild Cluster Bootstrap above (Cameron & Miller 2015).

1. Resample $G=8$ countries WITH replacement.
1. Resample $G=8$ countries WITH replacement.
2. Relabel resampled countries uniquely per draw (e.g. "SLV__0", "SLV__1") so that a country drawn more than once contributes as distinct panel entities with a valid (entity, time) index, and
2. Relabel resampled countries uniquely per draw (e.g. "SLV__0", "SLV__1") so that a country drawn more than once contributes as distinct panel entities with a valid (entity, time) index, and

so entity/time Two-Way FE are genuinely re-estimated on the resampled panel rather than reused from the original fit.
 so entity/time Two-Way FE are genuinely re-estimated on the resampled panel rather than reused from the original fit.

3. Refit EQ1, EQ2, EQ3 (identical Two-Way FE spec as Module 02) on the resampled panel; record β_1 , γ_1 , δ_1 and their product.
3. Refit EQ1, EQ2, EQ3 (identical Two-Way FE spec as Module 02) on the resampled panel; record β_1 , γ_1 , δ_1 and their product.

$B=500$ draws (lower than the $B=999$ wild bootstrap above because each draw now costs three PanelOLS fits instead of one).
 $B=500$ draws (lower than the $B=999$ wild bootstrap above because each draw now costs three PanelOLS fits instead of one).

Literature

Literature

Sobel (1982) — Asymptotic confidence intervals for indirect effects.

Sobel (1982) — Asymptotic confidence intervals for indirect effects.

MacKinnon, Lockhart & Hoffman (2004) — Confidence limits for the indirect effect: distribution of the product methods.
 MacKinnon, Lockhart & Hoffman (2004) — Confidence limits for the indirect effect: distribution of the product methods.

Preacher & Hayes (2008) — Asymptotic and resampling strategies for assessing and comparing indirect effects, *Behavior Research Methods*.
 Preacher & Hayes (2008) — Asymptotic and resampling strategies for assessing and comparing indirect effects, *Behavior Research Methods*.

Cameron & Miller (2015) — cluster-robust / cluster-bootstrap inference.
 Cameron & Miller (2015) — cluster-robust / cluster-bootstrap inference.

$b=100/500$ valid=100 fails=0

$b=100/500$ valid=100 fails=0

$b=200/500$ valid=200 fails=0

$b=200/500$ valid=200 fails=0

$b=300/500$ valid=300 fails=0

$b=300/500$ valid=300 fails=0

b=400/500 valid=400 fails=0
 b=400/500 valid=400 fails=0
 b=500/500 valid=500 fails=0
 b=500/500 valid=500 fails=0

Point estimates (original panel, no resampling):

Point estimates (original panel, no resampling):

beta1 (homicide_rate_log_lag1 → inst_avg) = -0.1809
 beta1 (homicide_rate_log_lag1 → inst_avg) = -0.1809
 gamma1 (inst_avg → fdi_percent_gdp) = 0.8524
 gamma1 (inst_avg → fdi_percent_gdp) = 0.8524
 delta1 (fdi_percent_gdp → gdp_growth) = 0.2015
 delta1 (fdi_percent_gdp → gdp_growth) = 0.2015
 indirect = beta1 × gamma1 × delta1 = -0.031081
 indirect = beta1 × gamma1 × delta1 = -0.031081

Cluster bootstrap over countries (B=500):

Cluster bootstrap over countries (B=500):

Mean(indirect*) = -0.030974 SD(indirect*) = 0.035551
 Mean(indirect*) = -0.030974 SD(indirect*) = 0.035551
 95% percentile CI = [-0.136968, 0.015011]
 95% percentile CI = [-0.136968, 0.015011]
 Bootstrap p-value (H0: indirect=0) = 0.1440
 Bootstrap p-value (H0: indirect=0) = 0.1440
 Valid draws: 500/500 | Fails: 0
 Valid draws: 500/500 | Fails: 0

■ 95% CI includes zero — the FULL three-stage chain is NOT statistically distinguishable from no effect at conventional levels.

■ 95% CI includes zero — the FULL three-stage chain is NOT statistically distinguishable from no effect at conventional levels.

This is the honest headline result for the 'mecanismo de transmisión' claim: each link is individually weak, and the chain as a whole does not clear a formal significance test either. Report the point estimate and CI, not a causal claim.

This is the honest headline result for the 'mecanismo de transmisión' claim: each link is individually weak, and the chain as a whole does not clear a formal significance test either. Report the point estimate and CI, not a causal claim.

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✓ Figure saved → figures\06b_mediation_bootstrap.png

✓ Figure saved → figures\06b_mediation_bootstrap.png

04-D — SE COMPARISON TABLE

04-D — SE COMPARISON TABLE

Equation Dep. var. Key var. β SE(CI.) p(CI.) SE(DK) SE(CR2) df_satt p(WCB)

Equation Dep. var. Key var. β SE(CI.) p(CI.) SE(DK) SE(CR2) df_satt p(WCB)

eq1 inst_avg homicide_rate_log_lag1 -0.1809 0.1070 0.093 0.0692 0.1305 2.16 0.085*

eq1 inst_avg homicide_rate_log_lag1 -0.1809 0.1070 0.093 0.0692 0.1305 2.16 0.085*

eq2 fdi_percent_gdp inst_avg 0.8524 0.2426 0.001 0.2825 0.2566 4.53 0.002***

eq2 fdi_percent_gdp inst_avg 0.8524 0.2426 0.001 0.2825 0.2566 4.53 0.002***

eq3 gdp_growth fdi_percent_gdp 0.2015 0.0772 0.010 0.1465 0.0790 2.87 0.108

eq3 gdp_growth fdi_percent_gdp 0.2015 0.0772 0.010 0.1465 0.0790 2.87 0.108

Columns:

Columns:

SE(Cl.) — conventional clustered SE [UNDERSIZED with G=8, use for reference only]
 SE(Cl.) — conventional clustered SE [UNDERSIZED with G=8, use for reference only]
 SE(DK) — Driscoll-Kraay HAC SE (robust to cross-sectional dependence)
 SE(DK) — Driscoll-Kraay HAC SE (robust to cross-sectional dependence)
 SE(CR2) — Bell-McCaffrey bias-corrected SE (Bell & McCaffrey 2002)
 SE(CR2) — Bell-McCaffrey bias-corrected SE (Bell & McCaffrey 2002)
 df(Satt.) — Satterthwaite effective df for t-test (Pustejovsky & Tipton 2018)
 df(Satt.) — Satterthwaite effective df for t-test (Pustejovsky & Tipton 2018)
 p(WCB) — wild cluster bootstrap p-value [PREFERRED for G=8]
 p(WCB) — wild cluster bootstrap p-value [PREFERRED for G=8]

Preferred inference: p(WCB) from wild cluster bootstrap (Webb weights).

Preferred inference: p(WCB) from wild cluster bootstrap (Webb weights).

SE(CR2) with df_Satterthwaite as secondary check.

SE(CR2) with df_Satterthwaite as secondary check.

WRITE-UP WARNING — do not cherry-pick p(Cl.):

WRITE-UP WARNING — do not cherry-pick p(Cl.):

p(Cl.) is the LEAST conservative column here by construction (this is

p(Cl.) is the LEAST conservative column here by construction (this is

exactly why Modules 04's bootstrap and CR2 exist in the first place —
 exactly why Modules 04's bootstrap and CR2 exist in the first place —
 see docstring, "Why not conventional clustered SE?"). Any claim in the
 see docstring, "Why not conventional clustered SE?"). Any claim in the
 paper that a coefficient is "robust across specifications" must be
 paper that a coefficient is "robust across specifications" must be
 checked against p(DK), p(CR2) AND p(WCB), not just p(Cl.). A table with
 checked against p(DK), p(CR2) AND p(WCB), not just p(Cl.). A table with
 all four is exported to tables/se_comparison.tex for direct inclusion
 all four is exported to tables/se_comparison.tex for direct inclusion
 in the manuscript.
 in the manuscript.

✓ LaTeX SE-comparison table saved → tables/se_comparison.tex (use this instead of quoting a single p-value in the manuscript)

✓ LaTeX SE-comparison table saved → tables/se_comparison.tex (use this instead of quoting a single p-value in the manuscript)

04-E — BOOTSTRAP FIGURES

04-E — BOOTSTRAP FIGURES

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Module 04 complete.

Module 04 complete.

✓ Module 04 completed in 509.4s

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RUNNING MODULE 05: 05_robustness.py

=====

05-A — LOAD

05-A — LOAD

✓ Data: 274 obs | 11 countries

✓ Data: 274 obs | 11 countries

ROBUSTNESS 1 — Leave-One-Country-Out (LOCO)

ROBUSTNESS 1 — Leave-One-Country-Out (LOCO)

Bias addressed: results driven by a single influential country.

Bias addressed: results driven by a single influential country.

EQ1: *inst_avg*

EQ1: *inst_avg*

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Full sample (baseline) 231 -0.1809 0.1070 0.093 * +0.0000

Full sample (baseline) 231 -0.1809 0.1070 0.093 * +0.0000

Excl. COL 208 -0.1173 0.0725 0.108 +0.0637

Excl. COL 208 -0.1173 0.0725 0.108 +0.0637

Excl. CRI 208 -0.2274 0.1113 0.043 ** -0.0465

Excl. CRI 208 -0.2274 0.1113 0.043 ** -0.0465

Excl. DOM 209 -0.1646 0.1012 0.106 +0.0163

Excl. DOM 209 -0.1646 0.1012 0.106 +0.0163

Excl. ECU 212 -0.1662 0.1263 0.190 +0.0147

Excl. ECU 212 -0.1662 0.1263 0.190 +0.0147

Excl. GTM 210 -0.1865 0.1117 0.097 * -0.0056

Excl. GTM 210 -0.1865 0.1117 0.097 * -0.0056

Excl. HND 208 -0.1967 0.1142 0.087 * -0.0158

Excl. HND 208 -0.1967 0.1142 0.087 * -0.0158

Excl. MEX 208 -0.1157 0.0796 0.148 +0.0653

Excl. MEX 208 -0.1157 0.0796 0.148 +0.0653

Excl. NIC 211 -0.2093 0.1108 0.061 * -0.0284

Excl. NIC 211 -0.2093 0.1108 0.061 * -0.0284

Excl. PAN 208 -0.1307 0.0854 0.128 +0.0502

Excl. PAN 208 -0.1307 0.0854 0.128 +0.0502

Excl. PER 220 -0.1888 0.1113 0.091 * -0.0078

Excl. PER 220 -0.1888 0.1113 0.091 * -0.0078

Excl. SLV 208 -0.3905 0.0942 0.000 *** -0.2096

Excl. SLV 208 -0.3905 0.0942 0.000 *** -0.2096

■ 0 sign reversal(s), 1 large shift(s) — check influential countries.

■ 0 sign reversal(s), 1 large shift(s) — check influential countries.

EQ2: *fdi_percent_gdp*

EQ2: *fdi_percent_gdp*

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Full sample (baseline) 264 0.8524 0.2426 0.001 *** +0.0000
 Full sample (baseline) 264 0.8524 0.2426 0.001 *** +0.0000
 Excl. COL 240 1.0164 0.4296 0.019 ** +0.1640
 Excl. COL 240 1.0164 0.4296 0.019 ** +0.1640
 Excl. CRI 240 0.8043 0.2615 0.002 *** -0.0482
 Excl. CRI 240 0.8043 0.2615 0.002 *** -0.0482
 Excl. DOM 240 0.7759 0.2537 0.003 *** -0.0765
 Excl. DOM 240 0.7759 0.2537 0.003 *** -0.0765
 Excl. ECU 240 0.7637 0.2766 0.006 *** -0.0888
 Excl. ECU 240 0.7637 0.2766 0.006 *** -0.0888
 Excl. GTM 240 0.8145 0.2440 0.001 *** -0.0379
 Excl. GTM 240 0.8145 0.2440 0.001 *** -0.0379
 Excl. HND 240 0.7685 0.2687 0.005 *** -0.0840
 Excl. HND 240 0.7685 0.2687 0.005 *** -0.0840
 Excl. MEX 240 0.7255 0.2603 0.006 *** -0.1270
 Excl. MEX 240 0.7255 0.2603 0.006 *** -0.1270
 Excl. NIC 240 0.9942 0.3116 0.002 *** +0.1417
 Excl. NIC 240 0.9942 0.3116 0.002 *** +0.1417

Excl. PAN 240 0.7432 0.3012 0.014 ** -0.1093
 Excl. PAN 240 0.7432 0.3012 0.014 ** -0.1093
 Excl. PER 240 0.9894 0.2472 0.000 *** +0.1370
 Excl. PER 240 0.9894 0.2472 0.000 *** +0.1370
 Excl. SLV 240 0.9163 0.2849 0.002 *** +0.0639
 Excl. SLV 240 0.9163 0.2849 0.002 *** +0.0639

✓ Robust: no sign reversals, no large shifts across LOCO.

✓ Robust: no sign reversals, no large shifts across LOCO.

EQ3: gdp_growth

EQ3: gdp_growth

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$



Full sample (baseline) 264 0.2015 0.0772 0.010 *** +0.0000
 Full sample (baseline) 264 0.2015 0.0772 0.010 *** +0.0000
 Excl. COL 240 0.1963 0.0787 0.013 ** -0.0052
 Excl. COL 240 0.1963 0.0787 0.013 ** -0.0052
 Excl. CRI 240 0.2054 0.0730 0.005 *** +0.0038
 Excl. CRI 240 0.2054 0.0730 0.005 *** +0.0038
 Excl. DOM 240 0.2283 0.0667 0.001 *** +0.0268
 Excl. DOM 240 0.2283 0.0667 0.001 *** +0.0268
 Excl. ECU 240 0.1629 0.0868 0.062 * -0.0386
 Excl. ECU 240 0.1629 0.0868 0.062 * -0.0386
 Excl. GTM 240 0.2240 0.0823 0.007 *** +0.0225
 Excl. GTM 240 0.2240 0.0823 0.007 *** +0.0225
 Excl. HND 240 0.2291 0.0708 0.001 *** +0.0276
 Excl. HND 240 0.2291 0.0708 0.001 *** +0.0276
 Excl. MEX 240 0.2068 0.0849 0.016 ** +0.0052
 Excl. MEX 240 0.2068 0.0849 0.016 ** +0.0052
 Excl. NIC 240 0.1856 0.0850 0.030 ** -0.0159
 Excl. NIC 240 0.1856 0.0850 0.030 ** -0.0159
 Excl. PAN 240 0.1324 0.1018 0.195 -0.0691
 Excl. PAN 240 0.1324 0.1018 0.195 -0.0691
 Excl. PER 240 0.1669 0.0919 0.071 * -0.0346

Excl. PER 240 0.1669 0.0919 0.071 * -0.0346
 Excl. SLV 240 0.2371 0.0885 0.008 *** +0.0356
 Excl. SLV 240 0.2371 0.0885 0.008 *** +0.0356
 ✓ Robust: no sign reversals, no large shifts across LOCO.
 ✓ Robust: no sign reversals, no large shifts across LOCO.

ROBUSTNESS 2 — Time Windows
 ROBUSTNESS 2 — Time Windows

Bias addressed: period-specific shocks, structural breaks.
 Bias addressed: period-specific shocks, structural breaks.

EQ1: inst_avg
 EQ1: inst_avg
 Specification N β SE p Stars $\Delta\beta$
 Specification N β SE p Stars $\Delta\beta$

Full sample 231 -0.1809 0.1070 0.093 * +0.0000
 Full sample 231 -0.1809 0.1070 0.093 * +0.0000
 Pre-2010 (pre-GFC) 76 -0.0233 0.1369 0.865 +0.1576
 Pre-2010 (pre-GFC) 76 -0.0233 0.1369 0.865 +0.1576
 Post-2010 155 -0.1242 0.0784 0.116 +0.0567
 Post-2010 155 -0.1242 0.0784 0.116 +0.0567
 Excl. COVID (2020–21) 210 -0.1530 0.0961 0.113 +0.0279
 Excl. COVID (2020–21) 210 -0.1530 0.0961 0.113 +0.0279
 Post-peace (2014+) 113 -0.1084 0.0733 0.143 +0.0725
 Post-peace (2014+) 113 -0.1084 0.0733 0.143 +0.0725

EQ2: fdi_percent_gdp
 EQ2: fdi_percent_gdp
 Specification N β SE p Stars $\Delta\beta$
 Specification N β SE p Stars $\Delta\beta$

Full sample 264 0.8524 0.2426 0.001 *** +0.0000
 Full sample 264 0.8524 0.2426 0.001 *** +0.0000
 Pre-2010 (pre-GFC) 99 2.1075 1.3477 0.122 +1.2551
 Pre-2010 (pre-GFC) 99 2.1075 1.3477 0.122 +1.2551
 Post-2010 165 1.5266 0.4487 0.001 *** +0.6741
 Post-2010 165 1.5266 0.4487 0.001 *** +0.6741
 Excl. COVID (2020–21) 242 0.9726 0.2887 0.001 *** +0.1201
 Excl. COVID (2020–21) 242 0.9726 0.2887 0.001 *** +0.1201
 Post-peace (2014+) 121 1.9035 0.5591 0.001 *** +1.0510
 Post-peace (2014+) 121 1.9035 0.5591 0.001 *** +1.0510

EQ3: gdp_growth
 EQ3: gdp_growth
 Specification N β SE p Stars $\Delta\beta$
 Specification N β SE p Stars $\Delta\beta$

```

Full sample 264 0.2015 0.0772 0.010 *** +0.0000
Full sample 264 0.2015 0.0772 0.010 *** +0.0000
Pre-2010 (pre-GFC) 99 0.1868 0.1240 0.136 -0.0147
Pre-2010 (pre-GFC) 99 0.1868 0.1240 0.136 -0.0147
Post-2010 165 0.2183 0.1500 0.148 +0.0167
Post-2010 165 0.2183 0.1500 0.148 +0.0167
Excl. COVID (2020–21) 242 0.1014 0.0707 0.153 -0.1001
Excl. COVID (2020–21) 242 0.1014 0.0707 0.153 -0.1001
Post-peace (2014+) 121 0.2080 0.2051 0.313 +0.0064
Post-peace (2014+) 121 0.2080 0.2051 0.313 +0.0064

```

ROBUSTNESS 3 — Alternative Lag Structures (EQ1: Violence → Institutions)

ROBUSTNESS 3 — Alternative Lag Structures (EQ1: Violence → Institutions)

Uses pre-computed lag columns from dataset (not re-computed).

Uses pre-computed lag columns from dataset (not re-computed).

Bias addressed: uncertainty about transmission delay of violence.

Bias addressed: uncertainty about transmission delay of violence.

EQ1: Alternative Lags

EQ1: Alternative Lags

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

```

Contemporaneous (lag 0) 232 -0.1164 0.1165 0.319 +0.0645
Contemporaneous (lag 0) 232 -0.1164 0.1165 0.319 +0.0645
Lag 1 (primary) 231 -0.1809 0.1070 0.093 * +0.0000
Lag 1 (primary) 231 -0.1809 0.1070 0.093 * +0.0000
Lag 2 232 -0.2946 0.1283 0.023 ** -0.1137
Lag 2 232 -0.2946 0.1283 0.023 ** -0.1137
Lag 3 223 -0.3660 0.1321 0.006 *** -0.1851
Lag 3 223 -0.3660 0.1321 0.006 *** -0.1851

```

ROBUSTNESS 3b — Lag Structure for EQ2 and EQ3

ROBUSTNESS 3b — Lag Structure for EQ2 and EQ3

Motivation: the theoretical model (paper Introduction) argues that security, institutions and growth adjust at DIFFERENT speeds -- institutions evolve gradually and growth responds with a further lag.

Motivation: the theoretical model (paper Introduction) argues that security, institutions and growth adjust at DIFFERENT speeds -- institutions evolve gradually and growth responds with a further lag.

Yet the PRIMARY specifications of EQ2 (institutions > FDI) and EQ3

Yet the PRIMARY specifications of EQ2 (institutions > FDI) and EQ3

(FDI + institutions -> growth) use CONTEMPORANEOUS regressors, which (FDI + institutions -> growth) use CONTEMPORANEOUS regressors, which is inconsistent with that narrative and leaves the door open to reverse causality / simultaneity (e.g. growth expectations attracting same-year FDI). This block re-estimates EQ2 and EQ3 replacing the contemporaneous institution/FDI regressors with their one-year lag (inst_avg_lag1, fdi_percent_gdp_lag1 -- built in Module 01) to test whether the results survive a specification that matches the stated theory.

EQ2: Contemporaneous vs Lagged Institutions

EQ2: Contemporaneous vs Lagged Institutions

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Contemporaneous inst_avg (primary) 264 0.8524 0.2426 0.001 *** +0.0000

Contemporaneous inst_avg (primary) 264 0.8524 0.2426 0.001 *** +0.0000

Lag 1: inst_avg_lag1 253 0.5698 0.2698 0.036 ** -0.2826

Lag 1: inst_avg_lag1 253 0.5698 0.2698 0.036 ** -0.2826

EQ3: Contemporaneous vs Lagged FDI+Institutions

EQ3: Contemporaneous vs Lagged FDI+Institutions

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Contemporaneous fdi+inst (primary) 264 0.2015 0.0772 0.010 *** +0.0000

Contemporaneous fdi+inst (primary) 264 0.2015 0.0772 0.010 *** +0.0000

Lag 1: fdi_lag1+inst_lag1 253 -0.0703 0.0587 0.233 -0.2718

Lag 1: fdi_lag1+inst_lag1 253 -0.0703 0.0587 0.233 -0.2718

■ Note: 'key' here is fdi_percent_gdp (or its lag) -- the coefficient printed. inst_avg's own coefficient in this lagged spec is in the JSON export, not this table.

■ Note: 'key' here is fdi_percent_gdp (or its lag) -- the coefficient printed. inst_avg's own coefficient in this lagged spec is in the JSON export, not this table.

ROBUSTNESS 4 — Alternative Institution Indices

ROBUSTNESS 4 — Alternative Institution Indices

Bias addressed: sensitivity to index construction methodology.

Bias addressed: sensitivity to index construction methodology.

EQ1: Alternative Institution Indices

EQ1: Alternative Institution Indices

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

inst_avg (primary) 231 -0.1809 0.1070 0.093 * +0.0000

inst_avg (primary) 231 -0.1809 0.1070 0.093 * +0.0000

inst_pca (PCA-based) 231 -0.3084 0.1817 0.091 * -0.1275

inst_pca (PCA-based) 231 -0.3084 0.1817 0.091 * -0.1275

rule_of_law 231 -1.9609 0.9337 0.037 ** -1.7800

rule_of_law 231 -1.9609 0.9337 0.037 ** -1.7800

control_corruption 231 -0.3954 0.8575 0.645 -0.2145

control_corruption 231 -0.3954 0.8575 0.645 -0.2145

political_stability 231 -2.6715 1.7807 0.135 -2.4906

political_stability 231 -2.6715 1.7807 0.135 -2.4906

✓ Sign consistent across all institution specifications.

✓ Sign consistent across all institution specifications.

ROBUSTNESS 5 — Alternative Control Sets

ROBUSTNESS 5 — Alternative Control Sets

Bias addressed: over-controlling or omitted variable bias.

Bias addressed: over-controlling or omitted variable bias.

■ 'Excl. GDP p.c. (bad-control check)': gdp_per_capita_log is plausibly

■ 'Excl. GDP p.c. (bad-control check)': gdp_per_capita_log is plausibly

DOWNSTREAM of institutions/FDI (part of the very causal channel this

DOWNSTREAM of institutions/FDI (part of the very causal channel this

pipeline studies), not an exogenous confounder. Including it as a

pipeline studies), not an exogenous confounder. Including it as a

control in EQ1/EQ2 risks a post-treatment ('bad control') bias that

control in EQ1/EQ2 risks a post-treatment ('bad control') bias that

partials out some of the effect of interest (Angrist & Pischke 2009,

partials out some of the effect of interest (Angrist & Pischke 2009,

ch. 3). This variant drops it while keeping the rest of the baseline

ch. 3). This variant drops it while keeping the rest of the baseline

controls, to see whether the key coefficient is sensitive to its

controls, to see whether the key coefficient is sensitive to its

inclusion.

inclusion.

EQ1: inst_avg

EQ1: inst_avg

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Baseline controls 231 -0.1809 0.1070 0.093 * +0.0000

Baseline controls 231 -0.1809 0.1070 0.093 * +0.0000

Extended controls 191 -0.2441 0.0732 0.001 *** -0.0632

Extended controls 191 -0.2441 0.0732 0.001 *** -0.0632

Minimal controls 231 -0.1935 0.1122 0.086 * -0.0126
 Minimal controls 231 -0.1935 0.1122 0.086 * -0.0126
 Excl. GDP p.c. (bad-control check) 231 -0.2091 0.1195 0.082 * -0.0281
 Excl. GDP p.c. (bad-control check) 231 -0.2091 0.1195 0.082 * -0.0281
 No controls (FE only) 231 -0.2545 0.1682 0.132 -0.0736
 No controls (FE only) 231 -0.2545 0.1682 0.132 -0.0736

EQ2: fdi_percent_gdp

EQ2: fdi_percent_gdp

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Baseline controls 264 0.8524 0.2426 0.001 *** +0.0000

Baseline controls 264 0.8524 0.2426 0.001 *** +0.0000

Extended controls 216 -0.4814 0.7636 0.529 -1.3339

Extended controls 216 -0.4814 0.7636 0.529 -1.3339

Minimal controls 264 0.4477 0.6373 0.483 -0.4047

Minimal controls 264 0.4477 0.6373 0.483 -0.4047

Excl. GDP p.c. (bad-control check) 264 0.8400 0.2319 0.000 *** -0.0124

Excl. GDP p.c. (bad-control check) 264 0.8400 0.2319 0.000 *** -0.0124

No controls (FE only) 264 -0.0315 0.4989 0.950 -0.8839

No controls (FE only) 264 -0.0315 0.4989 0.950 -0.8839

EQ3: gdp_growth

EQ3: gdp_growth

Specification N β SE p Stars $\Delta\beta$

Specification N β SE p Stars $\Delta\beta$

Baseline controls 274 0.2052 0.0862 0.018 ** +0.0000

Baseline controls 274 0.2052 0.0862 0.018 ** +0.0000

Extended controls 226 0.3650 0.1130 0.001 *** +0.1598

Extended controls 226 0.3650 0.1130 0.001 *** +0.1598

Minimal controls 274 0.2830 0.1130 0.013 ** +0.0778

Minimal controls 274 0.2830 0.1130 0.013 ** +0.0778

Excl. GDP p.c. (bad-control check) 274 0.2135 0.0798 0.008 *** +0.0083

Excl. GDP p.c. (bad-control check) 274 0.2135 0.0798 0.008 *** +0.0083

No controls (FE only) 274 0.2699 0.1083 0.013 ** +0.0647

No controls (FE only) 274 0.2699 0.1083 0.013 ** +0.0647

ROBUSTNESS 6 — Placebo Test (permuted homicide)

ROBUSTNESS 6 — Placebo Test (permuted homicide)

Procedure: randomly permute homicide_rate_log_lag1 within year cells.

Procedure: randomly permute homicide_rate_log_lag1 within year cells.

This breaks the country-specific causal link while preserving the

This breaks the country-specific causal link while preserving the

cross-sectional and temporal distribution of the variable.

cross-sectional and temporal distribution of the variable.

H0 (placebo): permuted violence has no effect on institutions.

H0 (placebo): permuted violence has no effect on institutions.

If the TRUE estimate is more extreme than 95% of placebo estimates,

If the TRUE estimate is more extreme than 95% of placebo estimates,

this confirms the result is not driven by spurious correlation.

this confirms the result is not driven by spurious correlation.

N_permutations = 500.

N_permutations = 500.

True estimate: $\beta_{\text{inst}} = -0.1809$

True estimate: $\beta_{\text{inst}} = -0.1809$

Placebo distribution: mean=0.0019, SD=0.0243, 5th pctl=-0.0388

Placebo distribution: mean=0.0019, SD=0.0243, 5th pctl=-0.0388

Placebo p-value: 0.0000

Placebo p-value: 0.0000

✓ Placebo test PASSED — true estimate more extreme than 95% of random permutations.

✓ Placebo test PASSED — true estimate more extreme than 95% of random permutations.

ROBUSTNESS 7 — Country-Specific Linear Trends

ROBUSTNESS 7 — Country-Specific Linear Trends

*Motivation: Two-Way FE (entity + time) removes country LEVELS and
Motivation: Two-Way FE (entity + time) removes country LEVELS and
common year shocks, but NOT country-specific trends. Several series
common year shocks, but NOT country-specific trends. Several series
in this panel are strongly trending over 25 years (GDP per capita,
in this panel are strongly trending over 25 years (GDP per capita,
population, and especially El Salvador's homicide collapse from >100
population, and especially El Salvador's homicide collapse from >100
to <10 per 100,000). If two trending series happen to move together
to <10 per 100,000). If two trending series happen to move together
for reasons unrelated to the hypothesized mechanism, Two-Way FE alone
for reasons unrelated to the hypothesized mechanism, Two-Way FE alone
will not catch it -- this is the classic 'spurious panel regression'
will not catch it -- this is the classic 'spurious panel regression'
risk (Granger & Newbold 1974; Kao 1999; Phillips & Moon 1999).
risk (Granger & Newbold 1974; Kao 1999; Phillips & Moon 1999).*

*This check re-estimates the PRIMARY specification of each equation
This check re-estimates the PRIMARY specification of each equation
adding one linear trend per country (country_dummy x year_c). If the
adding one linear trend per country (country_dummy x year_c). If the
key coefficient survives with the same sign and comparable magnitude,
key coefficient survives with the same sign and comparable magnitude,
the baseline result is not merely two coincidentally-trending series.
the baseline result is not merely two coincidentally-trending series.*

Reference: Wooldridge (2010), ch. 10.5 (unit-specific trends).

Reference: Wooldridge (2010), ch. 10.5 (unit-specific trends).

EQ1: inst_avg

EQ1: inst_avg

Specification N β_{inst} SE p Stars $\Delta\beta$

Specification N β_{inst} SE p Stars $\Delta\beta$

```

#####
#####
#####
#####
Baseline (no country trends) 231 -0.1809 0.1070 0.093 * +0.0000
Baseline (no country trends) 231 -0.1809 0.1070 0.093 * +0.0000
+ 11 country-specific trends 231 -0.1726 0.0803 0.033 ** +0.0083
+ 11 country-specific trends 231 -0.1726 0.0803 0.033 ** +0.0083
✓ Survives country-specific trends (Δ=5%, same sign).
✓ Survives country-specific trends (Δ=5%, same sign).

```

```

EQ2: fdi_percent_gdp
EQ2: fdi_percent_gdp
Specification N β SE p Stars Δβ
Specification N β SE p Stars Δβ
#####
#####
#####
#####

```

```

#####
#####
#####
#####
Baseline (no country trends) 264 0.8524 0.2426 0.001 *** +0.0000
Baseline (no country trends) 264 0.8524 0.2426 0.001 *** +0.0000
+ 11 country-specific trends 264 1.0457 0.5876 0.077 * +0.1932
+ 11 country-specific trends 264 1.0457 0.5876 0.077 * +0.1932
✓ Survives country-specific trends (Δ=23%, same sign).
✓ Survives country-specific trends (Δ=23%, same sign).

```

```

EQ3: gdp_growth
EQ3: gdp_growth
Specification N β SE p Stars Δβ
Specification N β SE p Stars Δβ
#####
#####
#####
#####

```

```

#####
#####
#####
#####
Baseline (no country trends) 264 0.2015 0.0772 0.010 *** +0.0000
Baseline (no country trends) 264 0.2015 0.0772 0.010 *** +0.0000
+ 11 country-specific trends 264 0.1811 0.1038 0.083 * -0.0205
+ 11 country-specific trends 264 0.1811 0.1038 0.083 * -0.0205
✓ Survives country-specific trends (Δ=10%, same sign).
✓ Survives country-specific trends (Δ=10%, same sign).

```

```

#####
#####
#####
#####
ROBUSTNESS 8 — Panel Unit Root Test (Fisher-ADF, Maddala & Wu 1999)
ROBUSTNESS 8 — Panel Unit Root Test (Fisher-ADF, Maddala & Wu 1999)
#####
#####

```

No unit-root test was previously implemented anywhere in this pipeline
 No unit-root test was previously implemented anywhere in this pipeline
 despite 25-year trending macro series. This runs an Augmented
 despite 25-year trending macro series. This runs an Augmented
 Dickey-Fuller test SEPARATELY for each country's time series (T=25 per
 Dickey-Fuller test SEPARATELY for each country's time series (T=25 per
 country, so maxlag=1, no automatic lag selection -- higher lags are
 country, so maxlag=1, no automatic lag selection -- higher lags are
 not identifiable with so few observations) and combines the p-values
 not identifiable with so few observations) and combines the p-values

with the Fisher (1932) / Maddala-Wu (1999) combination test:

with the Fisher (1932) / Maddala-Wu (1999) combination test:

$P = -2 * \sum(\log(p_i)) \sim \chi^2(2N)$ under H_0 : ALL panels have a unit root.

$P = -2 * \sum(\log(p_i)) \sim \chi^2(2N)$ under H_0 : ALL panels have a unit root.

CAVEAT (consistent with this pipeline's other small-sample warnings):

CAVEAT (consistent with this pipeline's other small-sample warnings):

with $T \sim 25$ observations per country, per-country ADF tests have very

with $T \sim 25$ observations per country, per-country ADF tests have very

low power. Failing to reject H_0 here is only weak evidence of a unit

low power. Failing to reject H_0 here is only weak evidence of a unit

root, not proof of one. This test should be read as a flag for

root, not proof of one. This test should be read as a flag for

further caution, not a definitive diagnosis -- see Robustness 7 above

further caution, not a definitive diagnosis -- see Robustness 7 above

(country-specific trends) as a partial, practical remedy regardless of

(country-specific trends) as a partial, practical remedy regardless of

the formal test outcome.

the formal test outcome.

Reference: Maddala & Wu (1999), A comparative study of unit root tests

Reference: Maddala & Wu (1999), A comparative study of unit root tests

with panel data and a new simple test, Oxford Bulletin of Economics

with panel data and a new simple test, Oxford Bulletin of Economics

and Statistics 61(S1), 631-652.

and Statistics 61(S1), 631-652.

Variable N countries Fisher χ^2 df p (Fisher) Flag

Variable N countries Fisher χ^2 df p (Fisher) Flag

homicide_rate_log 11 20.01 22 0.5825 ■ Cannot reject unit root (panel-wide) → spurious-regression risk

homicide_rate_log 11 20.01 22 0.5825 ■ Cannot reject unit root (panel-wide) → spurious-regression risk

inst_avg 11 32.15 22 0.0749 ■ Cannot reject unit root (panel-wide) → spurious-regression risk

inst_avg 11 32.15 22 0.0749 ■ Cannot reject unit root (panel-wide) → spurious-regression risk

fdi_percent_gdp 11 69.70 22 0.0000 ✓ Reject H_0 : at least one country's series is stationary

fdi_percent_gdp 11 69.70 22 0.0000 ✓ Reject H_0 : at least one country's series is stationary

gdp_growth 11 144.67 22 0.0000 ✓ Reject H_0 : at least one country's series is stationary

gdp_growth 11 144.67 22 0.0000 ✓ Reject H_0 : at least one country's series is stationary

gdp_per_capita_log 11 15.03 22 0.8607 ■ Cannot reject unit root (panel-wide) → spurious-regression risk

gdp_per_capita_log 11 15.03 22 0.8607 ■ Cannot reject unit root (panel-wide) → spurious-regression risk

05-G — FIGURES

05-G — FIGURES

✓ Figure saved → figures/07_robustness.png

✓ Figure saved → figures/07_robustness.png

05-H — EXPORT

05-H — EXPORT

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\05_robustness.json

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\05_robustness.json

Module 05 complete.

Module 05 complete.

✓ Module 05 completed in 71.3s

=====

RUNNING MODULE 06: 06_ml_triangular.py

=====

06-A — LOAD

06-A — LOAD

✓ Data: 274 obs | 11 countries

✓ Data: 274 obs | 11 countries

CV strategy: Leave-One-Country-Out (11 folds)

CV strategy: Leave-One-Country-Out (11 folds)

■ All ML results are EXPLORATORY. They do not imply causation.

■ All ML results are EXPLORATORY. They do not imply causation.

■ G=11 LOCO folds → CV performance estimates carry wide uncertainty. Interpret means alongside SDs, not in isolation.

■ G=11 LOCO folds → CV performance estimates carry wide uncertainty. Interpret means alongside SDs, not in isolation.

06-B — ML ESTIMATION (LOCO-CV, SHAP, ALE)

06-B — ML ESTIMATION (LOCO-CV, SHAP, ALE)

■■ T1: Violence → Institutions ■■

■■ T1: Violence → Institutions ■■

Missing summary:

Missing summary:

homicide_rate_log_lag2 0.153285

homicide_rate_log_lag2 0.153285

homicide_rate_log_lag1 0.120438

homicide_rate_log_lag1 0.120438

inst_avg 0.036496

inst_avg 0.036496

gdp_per_capita_log 0.000000

gdp_per_capita_log 0.000000

inflation 0.000000

inflation 0.000000

unemployment 0.000000

unemployment 0.000000

exports_percent_gdp 0.000000

exports_percent_gdp 0.000000

population_log 0.000000

population_log 0.000000

year_c 0.000000

year_c 0.000000

dtype: float64

dtype: float64

Rows before dropna: 274

Rows before dropna: 274

Rows after dropna: 226

Rows after dropna: 226

Model R^2 mean R^2 SD RMSE mean Valid folds

Model R^2 mean R^2 SD RMSE mean Valid folds

Random Forest -25.0716 51.6926 0.7465 11

Random Forest -25.0716 51.6926 0.7465 11

Gradient Boosting -38.0123 72.8938 0.8229 11

Gradient Boosting -38.0123 72.8938 0.8229 11

■ Note: G=11 LOCO folds → R^2 estimates have wide uncertainty. SD across folds is as informative as the mean.

■ Note: G=11 LOCO folds → R^2 estimates have wide uncertainty. SD across folds is as informative as the mean.

RF vs GB Wilcoxon signed-rank (paired LOCO R^2):

RF vs GB Wilcoxon signed-rank (paired LOCO R^2):

N pairs: 11 | W=33.0 p=1.0

N pairs: 11 | W=33.0 p=1.0

■ G=11 pairs → very low power. Treat p-value as descriptive only.

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RF OOB R^2 (internal diagnostic — in-sample, optimistic): 0.7645

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■ OOB R^2 is a training-data statistic, NOT a LOCO out-of-country R^2 . Do not cite OOB R^2 as evidence of predictive performance.

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Computing SHAP cluster bootstrap stability (B=200, cluster=country)...

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Feature Gini Perm±SD |SHAP| SHAP rank Rank SD P(top3) P(top5)

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Population (log) 0.3447 0.4090±0.0347 0.2405 1.40 0.97 0.94 0.99

Population (log) 0.3447 0.4090±0.0347 0.2405 1.40 0.97 0.94 0.99

GDP per capita (log) 0.1436 0.1329±0.0154 0.1226 3.33 1.59 0.58 0.89

GDP per capita (log) 0.1436 0.1329±0.0154 0.1226 3.33 1.59 0.58 0.89

Violence lag 2 0.1352 0.0907±0.0133 0.1193 2.81 1.12 0.77 0.96

Violence lag 2 0.1352 0.0907±0.0133 0.1193 2.81 1.12 0.77 0.96

Violence lag 1 0.1244 0.0676±0.0097 0.1087 3.96 1.15 0.41 0.90

Violence lag 1 0.1244 0.0676±0.0097 0.1087 3.96 1.15 0.41 0.90

Unemployment 0.0936 0.0688±0.0089 0.0729 5.06 1.59 0.14 0.66

Unemployment 0.0936 0.0688±0.0089 0.0729 5.06 1.59 0.14 0.66

Exports (% GDP) 0.0905 0.0553±0.0053 0.0410 5.17 1.54 0.14 0.45

Exports (% GDP) 0.0905 0.0553±0.0053 0.0410 5.17 1.54 0.14 0.45

Inflation 0.0497 0.0356±0.0037 0.0345 6.50 0.95 0.03 0.13

Inflation 0.0497 0.0356±0.0037 0.0345 6.50 0.95 0.03 0.13

Time Trend (centred) 0.0181 0.0120±0.0017 0.0182 7.79 0.65 0.01 0.02

Time Trend (centred) 0.0181 0.0120±0.0017 0.0182 7.79 0.65 0.01 0.02

Key feature 'homicide_rate_log_lag1':

Key feature 'homicide_rate_log_lag1':

Permutation rank: #5/8

Permutation rank: #5/8

SHAP rank: #4/8

SHAP rank: #4/8

■ Key feature does not rank in top 3. Inspect collinear features.

Computing SHAP cluster bootstrap stability (B=200, cluster=country)...

year_c 0.000000

[illegible]

Permutation rank: #2/8

SHAP rank: #1/8

SHAP rank: #1/8

✓ Key feature ranks in top 3 by both importance metrics.

✓ Key feature ranks in top 3 by both importance metrics.

06-C — CONVERGENCE TABLE: FE vs ML

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Convergence principle (Athey & Imbens 2019):

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If the variables that matter causally (significant in FE) also rank

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highly in ML importance, this triangulation strengthens the narrative.

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Divergence signals potential non-linearities or confounders.

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Task FE coef FE p-val FE sig Direction Perm rank SHAP rank Convergence

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T1: Violence → Institutions -0.1809 0.0926 * Neg 5 4 ■ Partial

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T2: Institutions → FDI 0.8524 0.0005 *** Pos 3 3 ✓ Strong

T2: Institutions → FDI 0.8524 0.0005 *** Pos 3 3 ✓ Strong

T3: FDI + Institutions → Growth 0.2015 0.0097 *** Pos 2 1 ✓ Strong

T3: FDI + Institutions → Growth 0.2015 0.0097 *** Pos 2 1 ✓ Strong

Notes:

Notes:

- FE coef and p-value are cluster-robust estimates from Module 02.

- FE coef and p-value are cluster-robust estimates from Module 02.

- FE sig: * p<0.10 ** p<0.05 *** p<0.01.

- FE sig: * p<0.10 ** p<0.05 *** p<0.01.

- Direction: sign of the FE coefficient for the key feature.

- Direction: sign of the FE coefficient for the key feature.

- Perm/SHAP rank: rank among all features in the ML model (1 = most important).

- Perm/SHAP rank: rank among all features in the ML model (1 = most important).

- Convergence: Strong = key feature ranks top 3 by both metrics.

- Convergence: Strong = key feature ranks top 3 by both metrics.

06-D — FIGURES

06-D — FIGURES

✓ Figure saved → figures/08_ml_loco_cv.png

✓ Figure saved → figures/08_ml_loco_cv.png

✓ Figure saved → figures/09_feature_importance.png

✓ Figure saved → figures/09_feature_importance.png

✓ Figure saved → figures/10_shap_ale.png

✓ Figure saved → figures/10_shap_ale.png

06-E — EXPORT

06-E — EXPORT

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✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\06_ml_results.json

Module 06 complete. Pipeline finished.

Module 06 complete. Pipeline finished.

■ IMPORTANT REMINDER ■

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■ ML results (SHAP, permutation importance, LOCO R^2) ■

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■ are EXPLORATORY and NON-CAUSAL. ■

■ are EXPLORATORY and NON-CAUSAL. ■

■ They triangulate — they do not identify causal effects. ■

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■ Primary causal evidence: Module 02 (Two-Way FE) and ■

■ Primary causal evidence: Module 02 (Two-Way FE) and ■

■ Module 04 (Wild Cluster Bootstrap inference). ■

■ Module 04 (Wild Cluster Bootstrap inference). ■

■ ■

■ ■

■ SHAP limitation: computed on the full-sample model. ■

■ SHAP limitation: computed on the full-sample model. ■

■ Values reflect in-sample feature associations and may not ■

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■ generalise to held-out countries. ■

■ generalise to held-out countries. ■

■ ■

■ ■

■ Bootstrap stability limitation: $G=8$ clusters → discrete ■

■ Bootstrap stability limitation: $G=8$ clusters → discrete ■

■ bootstrap distribution. Treat rank SDs and hit rates as ■

■ bootstrap distribution. Treat rank SDs and hit rates as ■

■ approximate lower bounds on true uncertainty. ■

■ approximate lower bounds on true uncertainty. ■

✓ Module 06 completed in 1265.0s

=====

RUNNING MODULE 07: 07_cointegration.py

=====

07-A — LOAD

07-A — LOAD

✓ Data loaded: 274 obs, 11 countries

✓ Data loaded: 274 obs, 11 countries

Scope reminder: this module only tests pairs of series that Module 05's

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Fisher-ADF test found to be $I(1)$ (non-stationary): homicide_rate_log,

Fisher-ADF test found to be $I(1)$ (non-stationary): homicide_rate_log,

inst_avg, gdp_per_capita_log, fdi_percent_gdp and gdp_growth already

inst_avg, gdp_per_capita_log, fdi_percent_gdp and gdp_growth already

reject a unit root there (they are $I(0)$) and are intentionally excluded

reject a unit root there (they are $I(0)$) and are intentionally excluded

from cointegration testing -- see module docstring.

from cointegration testing -- see module docstring.

07-B — STEP 1: LONG-RUN RELATIONSHIP & RESIDUAL COINTEGRATION TEST

07-B — STEP 1: LONG-RUN RELATIONSHIP & RESIDUAL COINTEGRATION TEST

■ ■ T1: Homicide & Institutions ■ ■

■ ■ T1: Homicide & Institutions ■ ■

Long-run regression: $inst_avg_it = a_i + I_t + \beta * homicide_rate_log_it + e_{it}$

Long-run regression: $inst_avg_it = a_i + I_t + \beta * homicide_rate_log_it + e_{it}$

$\beta = -0.2415$ SE(cl) = 0.1886 p(cl) = 0.2017 N = 232

$\beta = -0.2415$ SE(cl) = 0.1886 p(cl) = 0.2017 N = 232

Fisher-ADF on residuals: $\chi^2(22)=22.74$, $p=0.4167$

Fisher-ADF on residuals: $\chi^2(22)=22.74$, $p=0.4167$

■ Cannot reject unit root in residuals ($p=0.4167$) -& NO evidence of cointegration; levels relationship may be spurious

■ Cannot reject unit root in residuals ($p=0.4167$) -& NO evidence of cointegration; levels relationship may be spurious

■ ■ T2: Institutions & Income ■ ■

■ ■ T2: Institutions & Income ■ ■

Long-run regression: $inst_avg_it = a_i + I_t + \beta * gdp_per_capita_log_it + e_{it}$

Long-run regression: $inst_avg_it = a_i + I_t + \beta * gdp_per_capita_log_it + e_{it}$

$\beta = 0.9651$ SE(cl) = 0.3942 p(cl) = 0.0151 ** N = 264

$\beta = 0.9651$ SE(cl) = 0.3942 p(cl) = 0.0151 ** N = 264

Fisher-ADF on residuals: $\chi^2(22)=30.18$, $p=0.1141$

Fisher-ADF on residuals: $\chi^2(22)=30.18$, $p=0.1141$

■ Cannot reject unit root in residuals ($p=0.1141$) -& NO evidence of cointegration; levels relationship may be spurious

■ Cannot reject unit root in residuals ($p=0.1141$) -& NO evidence of cointegration; levels relationship may be spurious

■ ■ T3: Homicide & Income ■ ■

■ ■ T3: Homicide & Income ■ ■

Long-run regression: $gdp_per_capita_log_it = a_i + I_t + \beta * homicide_rate_log_it + e_{it}$

Long-run regression: $gdp_per_capita_log_it = a_i + I_t + \beta * homicide_rate_log_it + e_{it}$

$\beta = -0.0911$ SE(cl) = 0.0962 p(cl) = 0.3447 N = 242

$\beta = -0.0911$ SE(cl) = 0.0962 p(cl) = 0.3447 N = 242

Fisher-ADF on residuals: $\chi^2(22)=39.94$, $p=0.0110$

Fisher-ADF on residuals: $\chi^2(22)=39.94$, $p=0.0110$

✓ REJECT unit root in residuals ($p=0.0110$) -& evidence of cointegration

✓ REJECT unit root in residuals ($p=0.0110$) -& evidence of cointegration

07-C — STEP 2: ERROR-CORRECTION MODEL (cointegrated pairs only)

07-C — STEP 2: ERROR-CORRECTION MODEL (cointegrated pairs only)

■■ T1: Homicide &-> Institutions ■■

■■ T1: Homicide &-> Institutions ■■

■ Not cointegrated in Step 1 -- ECM skipped (would be a regression of one I(1)-differenced series on an unreliable, non-mean-reverting 'equilibrium error' -- not meaningful).

■ Not cointegrated in Step 1 -- ECM skipped (would be a regression of one $I(1)$ -differenced series on an unreliable, non-mean-reverting 'equilibrium error' -- not meaningful).

■■ T2: Institutions &-& Income ■■

■■ T2: Institutions &-> Income ■■

■ Not cointegrated in Step 1 -- ECM skipped (would be a regression of one I(1)-differenced series on an unreliable, non-mean-reverting 'equilibrium error' -- not meaningful).

■ Not cointegrated in Step 1 -- ECM skipped (would be a regression of one I(1)-differenced series on an unreliable, non-mean-reverting 'equilibrium error' -- not meaningful).

■■ T3: Homicide &-& Income ■■

■■ T3: Homicide & Income ■■

$$\Delta \ln gdp_per_capita_it = a_i + \lambda_t + \phi_i \hat{e}_{it-1} + \gamma_i \Delta \ln homicide_rate_it + \epsilon_{it}$$
$$\Delta \ln gdp_per_capita_it = a_i + \lambda_t + \phi_i \hat{e}_{it-1} + \gamma \Delta \ln homicide_rate_it + \epsilon_{it}$$

ϕ (speed of adjustment) = -0.1092 SE(cI)=0.0391 p(cI)=0.0058 p(dk)=0.0000 ***

$$\text{phi (speed of adjustment)} = -0.1092 \text{ SE(cl)}=0.0391 \text{ p(cl)}=0.0058 \text{ p(dk)}=0.0000 \text{ ***}$$

γ (short-run effect) = -0.0333 SE(cI)=0.0351 p(cI)=0.3440 p(dk)=0.1855

γ (short-run effect) = -0.0333 SE(cI)=0.0351 p(cI)=0.3440 p(dk)=0.1855

$N = 227$ Within $R^2 = 0.0265$

N = 227 Within $R^2 = 0.0265$

✓ $\phi \neq 0$ and significant: $\text{gdp_per_capita_log}$ corrects 10.9% of a deviation from long-run equilibrium with homicide_rate_log each year.

✓ $\phi \neq 0$ and significant: $\text{gdp_per_capita_log}$ corrects 10.9% of a deviation from long-run equilibrium with homicide_rate_log each year.

07-D — FIGURES

07-D — FIGURES

✓ Figure saved -> figures/11_cointegration_residuals.png

✓ Figure saved -> figures/11_cointegration_residuals.png

07-E — EXPORT

07-E — EXPORT

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\07_cointegration.json

✓ Saved → C:\data_analyzer\econometric_pipeline\pipeline\json\07_cointegration.json

Module 07 complete.

Module 07 complete.

✓ Module 07 completed in 9.9s

PIPELINE SUMMARY

✓ Module 01: Data Preparation & Institution Index

✓ Module 02: Panel Estimation (Two-Way FE)

✓ *Module 03: Panel Diagnostics*

✓ *Module 04: Wild Cluster Bootstrap Inference*

✓ Module 05: Robustness Checks

✓ Module 06: ML Triangulation

✓ Module 07: Panel Cointegration & ECM

Total runtime: 1899.9s

Output directories:

Figures → C:\data_analyzer\econometric_pipeline\pipeline\figures/

Tables → C:\data_analyzer\econometric_pipeline\pipeline\tables/

JSON → C:\data_analyzer\econometric_pipeline\pipeline\json/

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